
Statistic Preservation Is Not Security Preservation: Identification Limits of Outcome-Gated Screening for Federated-Learning Defense Composition

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Abstract

Composing federated-learning defenses invites a failure mode: upstream processing can destroy the property the downstream defense reads. The natural repair—require the upstream stage to **preserve** it—is **the wrong abstraction for security**, and we establish that by intervention, not contrast. Preservation has five **levels**: the statistic’s value, the ordering it induces, the decision that follows, the adversarial mass admitted, and suppression. The first three chain by definition; **below the decision the chain breaks in both directions**. With the downstream defense, the attack, and standalone effectiveness held fixed, an oracle pinning every adversary’s coefficient—closing the attenuation channel by construction—isolates statistic disturbance from it, and **that disturbance leaves suppression inside our pre-specified margin**: Krum’s decision flips in up to 80% of rounds while the adversarial mass it admits stays exactly 0.000. In the same cell the effect **reverses sign** once adversarial influence co-varies: the outcome-gated ladder *falls* 0.272 where the instrument *rises* 0.098. An **identification boundary** explains why: a criterion gated on the outcome it predicts conditions on a descendant of mechanism and outcome, so **no cross-defense contrast identifies mechanism’s contribution at any sample size**—only within-defense intervention does. We classify which downstream signals survive the positive per-client rescalings defenses apply, prove sufficient conditions for the weighting→aggregator class, and prove **statistic-level non-identifiability** and that dispersion and adversary-correlated targeting are independently controllable. Three predictors are refuted, two by our own pre-registered tests. **The negative rests on oracle intervention on two arms, not a general law.**

1 Introduction

Federated learning [McMahan et al., 2017] enables collaborative model development across distributed compute, but operating in trustless environments—where participants cannot be vetted—introduces vulnerability to data poisoning [Chen et al., 2017, Bagdasaryan et al., 2020]. An arms race has produced defenses such as FoolsGold [Fung et al., 2020], RFA [Pillutla et al., 2022], Trimmed Mean [Yin et al., 2018], and Coordinate-wise Median [Blanchard et al., 2017], each imposing accuracy/robustness trade-offs with none universally dominant.

A natural question: *can we compose multiple defenses for stronger robustness?* The answer is non-trivial, and the cleanest illustration is that composition does not commute. FoolsGold→CoordMedian suppresses a pixel backdoor to ASR 0.076; reversing the order raises it to 0.732, because a rank aggregator emits a single vector rather than per-client updates and so

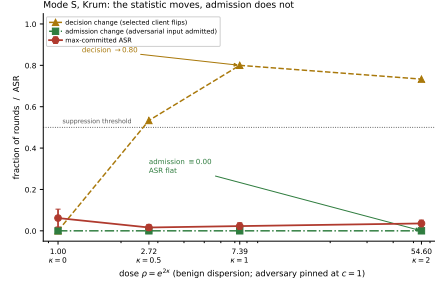


Figure 1: **The paper in one figure: the statistic moves, admission does not, and suppression holds.** Mode S, Krum under model-scaling (§4). Every adversary is pinned at $c=1$, so the adversarial coefficient share is constant across rungs, Lemma 1’s attenuation channel is shut *by construction*, and the dose **isolates statistic disturbance from adversarial coefficient attenuation**—the one channel it closes, and not the only thing it moves. As benign dispersion ρ grows to 54.6, Krum’s *decision* flips in up to 80% of rounds while *admission* of adversarial input holds at exactly 0.000 and max-committed ASR shows no evidence of the predicted increase, staying inside the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$) and low. **Statistic preservation is therefore not what was doing the work.** Bars are Student- t 95% intervals ($n=5$); every value recomputed per seed by `analyze_targeted_dose.py`. A pre-registered replication in a *different* invariance class and on a *second dataset and architecture* reproduces the negative and **refutes** the cross-arm admission ordering we had predicted (§5.3).

cannot serve as an *upstream* stage at all: the pair collapses to FoolsGold alone, bit-identical to FoolsGold with no upstream defense. The discriminative signal is not weakened; the composition ceases to exist. Naive layering fails this way whenever upstream processing destroys what the downstream defense reads, and randomizing which defense runs each round inherits the problem rather than escaping it.

Both naive approaches fail, and the natural repair—check that the upstream stage preserves the property the downstream defense reads—is the subject of this paper. **One thesis governs everything below: downstream-statistic preservation is causally insufficient for attack-suppression preservation, and the screening rule is a consequence of that, not the contribution.** With the measurement attached: with standalone effectiveness controlled and the intervention applied *directly* to the statistic the downstream defense reads, changing that statistic can have little or no effect on suppression—and the effect *reverses sign* once the intervention lets adversarial influence co-vary instead (-0.272 co-varying, $+0.098$ pinned, §4). *Security preservation* names that thesis’s object throughout—suppression of the committed attack surviving composition, under the threat model below—and is never conflated with the two levels it decomposes into, *adversarial contribution* (the influence the adversary retains on the aggregate) and *attack-suppression preservation* (ASR below threshold); none of the three ever means preservation of the statistic, the ordering it induces, or the decision that follows. **The second half of the contribution is the experimental design that can test a composition mechanism at all**, which follows from an **identification boundary**: any criterion whose eligibility condition is defined on the **same outcome metric it predicts** cannot be validated by cross-defense contrasts, because the gate forces effectiveness and mechanism to co-vary across every certified candidate—so **the admissible designs are within-defense**. One scoping point that sharpens rather than weakens the claim: C1 requires that some constituent already suppress the attack alone, so nothing here discovers defenses from nothing—it reasons about *compositions of already-effective defenses*.

The result, up front. Preservation is a *hierarchy*, not a property: the statistic’s value (P1), the ordering it induces (P2), the decision that follows (P3), the adversarial mass admitted (P4), suppression itself (P5). We show $(P1) \Rightarrow (P2) \Rightarrow (P3) \not\Rightarrow (P4) \not\Rightarrow (P5)$, the first two links holding **by definition** and each of the last two broken by a **counterexample we measured**. $(P3) \not\Rightarrow (P4)$: Mode S sweeps benign dispersion ρ from 1 to 54.6 and flips Krum’s decision in up to 80% of rounds while the adversarial coefficient share stays pinned at exactly 0.266667, admitted mass at 0.000, and ASR inside the pre-specified equivalence margin (Figure 1). $(P4) \not\Rightarrow (P5)$: `cos_krum` holds statistic, ordering, decision and admission *bit-identical* at every rung and its ASR falls 0.425 regardless. Hence, formally (Proposition 4, §4): **no level of this hierarchy is sufficient for security**—a non-existence result *within* the positive-rescaling class on these defenses, exhibited by counterexample, not a claim about every conceivable metric. **Everything an invariance argument can certify lives at (P1)–(P3); the security content is at (P4)**, and that gap—not the screening rule built over it—is this paper’s subject.

1. **The preservation hierarchy, and the two-directional break above.** Because the per-client defenses in use are all positive scalar rescalings, mechanism preservation is a decidable *invariance* question (Proposition 1), and on the weighting $\rightarrow$ aggregator class we prove it with explicit margins (Theorem 1)—a **boundary** theorem, not a security theorem. Definition 2 gives the five levels the box states, and §4 measures both breaks. One corollary is worth stating alone: weight heterogeneity is safe at *both* extremes and dangerous in the middle, so **the weight ratio ρ alone mispredicts** (Lemma 1, §2.2).
2. **An identification boundary for outcome-gated screening, and the design class it leaves admissible.** Proposition 2 proves the general claim above; its FL instance is that $C0 \wedge C1$ force the downstream defense of every certified pair to suppress the attack alone—**mechanism-effectiveness confounding** (§3)—and its evidential cost is real: a matched ablation built to break the confound had its premise refuted by its own pre-freeze baselines, so **the confound re-materialized inside the experiment designed to remove it**. A second boundary limits coverage rather than validation: **emergent** suppression—present in the pair, absent in both constituents—cannot satisfy an inherited-suppression gate, capping such a screen’s recall (Proposition 3), and our own best composition is its witness.
3. **Three central empirical findings, all causal interventions—not a validation of any screen.** (i) *The negative, instrumented and replicated (Causal, then Replication).* Mode S pins every adversary’s coefficient, closing Lemma 1’s attenuation channel *by construction*; a **pre-registered replication** in a *different* invariance class and on a *second dataset and architecture* reproduces it while **refuting the cross-arm admission ordering we had predicted** ($R^2=0.239$ over 26 cells), so **admission is a direction to measure, not a quantitative predictor**. (ii) *A sign reversal under a change of intervention (Causal).* The same cell yields -0.272 ($p_{\downarrow}=0.0015$) with adversarial influence free to co-vary and $+0.098$ with it pinned: the confounded design does not lose power, **it gets the sign wrong**—hence we withdraw the pooled form of C2’s causal claim (80 runs; $\rho_s = +0.351$, $p = 0.091$; §5.2). (iii) *Heterogeneity closes the theorem’s own margin.* At Dirichlet $\alpha=0.1$ RFA’s all-round mean separation margin collapses from $m=3.24$ to $m=1.01$ ($4.71 \rightarrow 2.06$ over rounds with $w_a > 0$), the limit at which Corollary 1’s bound goes vacuous, and two of three certified compositions cross the threshold, so the guarantee is measurably empty where heterogeneity is high (§6). A fourth predictor falls with them: **adversary-correlated dispersion** Δ_c is independently controllable from ρ (Proposition 5) and locates the two real anchors our synthetic curve misses, but **orders suppression no better**. Statistic disturbance, admission magnitude and coefficient targeting are each refuted *as a predictor*: what survives is a preservation *level*, not a statistic of the coefficient vector.
4. **The screening rule is a consequence of (2), not a claim of its own** (*Screening* throughout). What survives the negative is a conservative *evaluation-prioritization* rule: read C2 off the downstream signal’s invariance class, then evaluate the survivors first. We report it as an application (§6), never as a validated security predictor—it certifies only *inherited* robustness under a *committed* attack, at recall 40%, and outputs an evaluation *ordering*, never a deployment decision.
5. **Secondary results.** Fixed-order composition beats randomized ordering $9\times$, because preservation conditions are order-sensitive; and **one** whitebox composition-aware *adaptive* stress test raises $FG \rightarrow RFA$ to 0.167 ± 0.069 , $3.3\times$ its seed-matched committed 0.051—**an inherited-robustness guarantee is not adaptive robustness** (§8, Appendix K).

104 **Three tiers of evidence, of very unequal weight, never pooled.** Every empirical claim about the
 105 preservation hierarchy or the criterion is tagged with one of these three words where it is made. The
 106 secondary results carry no tag: neither the ordering comparison nor the *single* adaptive stress test is
 107 offered as evidence about the hierarchy, and the adaptive test is one test on one composition.

	tier	design	what it licenses
	Causal	within-defense intervention at fixed d_2 , attack and C1 input: Mode S, Mode A (§4)	statements about the <i>mechanism</i> , scoped to these cells
108	Replication	FEMNIST—second dataset <i>and</i> architecture ($\Delta = -0.017$)—and <code>coord_median</code> , a second invariance class (+0.098)	“replicated on a second dataset and architecture”, never “generalizes”
	Screening	the outcome-gated 42-pair suite, 38/42	evaluation prioritization only ; by Proposition 2 it cannot identify the mechanism’s contribution at any sample size

109 **How the paper is organized, as an argument.** Composition creates an **identification problem**.
 110 §2 defines the object to be identified—mechanism preservation as invariance, the five levels, the
 111 conditions; §3 characterizes the **boundary** on identifying it and *derives* the class of designs that
 112 can; §4 runs the experiment that class dictates; §5 reports what the boundary **falsifies**, our own
 113 pooled hypothesis included; §6 states what survives.

114 **Why this class.** Everything we *prove* is instantiated for positive per-client rescalings, deliberately:
 115 the class is a **laboratory for studying composition, not a general model of it**—it covers the per-
 116 client defenses in current use (norm clipping, FoolsGold, RFA reweighting, FLTrust) and makes
 117 invariance decidable from definitions. **Every claim here is general, formal or empirical, and they**
 118 **do not transfer upward.** Table 1, first in the appendix, is the one place each scope condition is
 119 stated—what falls outside the class, what the theorems do and do not bound, the sample sizes, what
 120 “refuted” means, why our instruments are not deployable—and we do not restate them at each use.

121 **Threat model and scope.** We consider persistent backdoor attacks where an adversary controlling
 122 fraction f of clients embeds triggers that survive across rounds, and define *persistence* γ as the frac-
 123 tion of backdoor ASR retained per round of clean aggregation ($\gamma \approx 0.66$ for CIFAR-10/CNN). At-
 124 tacks include model-scaling [Bagdasaryan et al., 2020], pixel-pattern backdoors [Chen et al., 2017],
 125 distributed backdoor attacks [Xie et al., 2020], and adaptive composition-aware adversaries [She-
 126 jwalkar and Houmansadr, 2021]. The framework constrains the aggregation pipeline, not the com-
 127 munication graph, and is evaluated only in **star-topology FL**. **A scope condition on the criterion**
 128 **itself, not a limitation of our runs:** C1 and every prediction built on it are defined against a *com-*
 129 *mitted* adversary, so what is bounded is *inherited* robustness under a fixed attack, **never adaptive**
 130 **robustness** (Table 1).

131 2 Mechanism Preservation: Invariance Classes

132 **Definition 1** (Defense composition). *Given defenses d_1 and d_2 , composition ($d_1 \rightarrow d_2$) operates*
 133 *as: d_1 produces per-client transformed updates $U' = \{T_{d_1}(u_i)\}_{i=1}^K$; d_2 aggregates U' using its*
 134 *standard algorithm. For FoolsGold $\rightarrow$ CoordMedian, FG outputs $\{w_i u_i\}$ and CoordMedian takes*
 135 *the unweighted coordinate-wise median of the K transformed values, not a “weighted median”.*
 136 *Clients with $w_i = 0$ are **not removed** but reach d_2 as zero vectors, so at least one point of U' is*
 137 *always the origin—which is what Lemma 1 is about.*

138 Each defense suppresses malicious updates through a *discriminative property*—a structural feature
 139 separating adversarial from benign contributions—and the property is defense-class-specific: *simi-*
 140 *larity spread* (FoolsGold), *coordinate ordering* (TrimmedMean, CoordMedian), *residual magnitude*
 141 (RFA), *consensus distance* (Reputation, in ℓ_2 from the coordinate-wise median) and *norm magni-*
 142 *tude* (NormClip). Mechanism preservation is then governed by whether d_1 ’s transform leaves d_2 ’s
 143 property intact, and because every per-client defense we consider acts as a positive scalar rescaling,
 144 that question has a sharp answer:

145 **Proposition 1** (Invariance classes under positive per-client rescaling). *Let $T(u_i) = c_i u_i$ with $c_i > 0$,
 146 **strictly**—the form taken by NormClip ($c_i = \min(1, \tau/\|u_i\|)$), Reputation ($c_i \propto e^{-d_i/s}$) and RFA-
 147 reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \epsilon)$), each positive by construction. **FoolsGold is deliberately**
 148 **not in that list**: its shift-and-clamp gives the least-trusted client weight exactly 0 in every round (2 of
 149 5 clients here), so FoolsGold-as- d_1 falls outside the hypothesis and is governed by Lemma 1 rather
 150 than by (a)–(c). Then:*

- 151 (a) **Cosine similarity is exactly invariant**: $\cos(c_i u_i, c_j u_j) = \cos(u_i, u_j)$. Hence FoolsGold’s
 152 weight vector is unchanged, $w(T(U)) = w(U)$, and **C2 holds for every $\cdot \rightarrow$ FoolsGold**
 153 **composition**.
- 154 (b) **Coordinate ordering is not invariant** in general; it survives under the bounded-ratio con-
 155 dition of Theorem 1(1).
- 156 (c) **Consensus distance and residual magnitude are not invariant**: heterogeneous c_i can in-
 157 vert their ordering, so $\cdot \rightarrow$ Reputation compositions are C2 failures and $\cdot \rightarrow$ RFA requires
 158 the separation condition of Theorem 1(2).

159 So mechanism preservation is a question about the *invariance class* of the downstream signal, and
 160 the classification corrects a natural but wrong intuition: norm clipping does not blind FoolsGold,
 161 though it does blind Reputation, whose signal is consensus distance rather than similarity.

162 **The framework in general form.** The structure does not depend on rescaling. Write d_2 ’s dis-
 163 criminative property as a statistic $S(U)$ and let T be d_1 ’s per-client transform; then $(d_1 \rightarrow d_2)$ is
 164 **mechanism-preserving** when $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering d_2 acts on.
 165 C2 is exactly that requirement, C3 asks in addition that φ not compress the gap below d_2 ’s operating
 166 threshold, and Proposition 1 is the case $T(u_i) = c_i u_i$ (Appendix G)—so the conditions are alge-
 167 braic rather than experimental. It also states the real limit—a d_1 **whose transform is not per-client**
 168 (mixing updates, or carrying cross-round state) **has no T of this form, and nothing here applies**.

169 Writing C2 as an invariance of S leaves a gap our own experiments later exposed, so we name the
 170 levels before stating the condition. They are the paper’s main conceptual instrument, and every
 171 claim below is stated at one of them:

172 **Definition 2** (Preservation levels). *For an upstream transform T and a downstream defense d_2*
 173 *reading statistic S :*

- 174 (P1) **Value invariance**: $S(T(U)) = S(U)$.
- 175 (P2) **Ordering invariance**: $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering S induces
 176 on clients. This is the level C2 is stated at, and the distinction from (P1) matters: exact
 177 values are sufficient but never necessary.
- 178 (P3) **Decision invariance**: the client or coordinate d_2 actually selects is unchanged.
- 179 (P4) **Adversarial contribution (admission) invariance**: the adversarial input d_2 admits into its
 180 aggregate is unchanged. This is a **level, not one measurable quantity**: for a selector it
 181 is the indicator that an adversarial update is selected at all; for a weighted averager, the
 182 adversarial share of the weight mass; for a coordinate-wise order statistic, the adversarial
 183 share of selected coordinates—which is why the influence column of Appendix B’s table is
 184 aggregator-specific by necessity. The committed pre-registrations say “admission”, so that
 185 word is retained in every frozen rule name and no scored quantity is relabelled.
- 186 (P5) **Attack-suppression preservation**: ASR stays below threshold.

187 (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) **hold by definition—logical implications, not empirical claims**; their content is
 188 that they stop at (P3), and §1’s box states both breaks below it. Holding decision and admission
 189 invariance does not deliver suppression: *cos.krum* falls 0.173 by the first rung alone through the
 190 magnitude channel—at $p=2.72$, where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy
 191 gate passes (paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). Across the whole ladder it
 192 falls 0.425, but that endpoint sits below our accuracy floor, so **the gate-passing rung is the witness**
 193 **we rest on** (§5.3). Losing decision invariance does not cost suppression: Krum’s selection flips in
 194 up to 80% of rounds under an intervention pinning the adversarial mass, and its ASR stays inside
 195 the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$) (§4). So a criterion stated at (P1)–(P2)
 196 constrains security in neither direction, (P4), where the security content sits, is not sufficient for
 197 (P5) either, and the hierarchy answers “**which level does a security claim have to be stated at?**”,
 198 not “**which level predicts ASR?**”

199 **The full causal chain, and the links we actually measure.** Definition 2 is a prefix of a longer
200 chain: *statistic* $\rightarrow$ *decision* $\rightarrow$ *admission* $\rightarrow$ *aggregate adversarial influence* $\rightarrow$ per-round model
201 update $\rightarrow$ attack persistence $\rightarrow$ ASR. We measure the four emphasized links and **not** per-round
202 update geometry or persistence dynamics—an acknowledged gap, not an assumed-benign one (Fig-
203 ure 2). A selection flipping from one benign client to another disturbs the statistic maximally and
204 the mechanism not at all, which is why (P3) and (P4) come apart. **“Mechanism” always means the**
205 **downstream defense’s discriminative operation, never its security behaviour**, since that impli-
206 cation is what this paper refutes; “mechanism preservation” is reserved for the property Theorem 1
207 establishes, and we say *statistic* or *decision* preservation where only the weaker property holds.

208 **Definition 3** (Signal-Preservation Conditions). *Composition* $(d_1 \rightarrow d_2)$ is predicted to suppress
209 attack a when: **(C0)** d_2 ’s mechanism is load-bearing, $\text{ASR}(d_1, a) \geq 0.5$ for some committed a —a
210 condition on the criterion’s domain that **never certifies a pair**, and one we formulated only after
211 §5.1 made it bind, so it is applied **retrospectively** and is **not part of what the 42 pairs validate**;
212 **(C1)** some constituent suppresses every committed attack alone, $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$ for
213 all a —defined through the same threshold that constitutes the outcome, hence an **empirical prereq-**
214 **uisite and not a mechanistic property**; **(C2)** d_1 ’s transform does not destroy d_2 ’s discriminative
215 property, decided from definitions alone by Proposition 1; **(C3)** it does not compress the quantitative
216 margin below d_2 ’s operating threshold. **The prioritization rule is $C1 \wedge C2$, C3 demoted to a margin**
217 **diagnostic** (Appendix Q); the pre-registered labels of §5 were frozen with C3 in the gate and are not
218 relabelled. **Theorem 1 bounds no ASR:** that mechanism preservation plus C1 predicts low ASR is a
219 separate empirical hypothesis, tested in §5, and **neither stage alone establishes low ASR, nor does**
220 **their conjunction** (stage structure, Appendix A).

221 2.1 Formal Guarantee: Weighting $\rightarrow$ Aggregator Class

222 For the practically important class containing our certified compositions—a per-client *weighting*
223 defense d_1 followed by a *rank-based* or *geometric* aggregator d_2 —C2 and C3 can be formally es-
224 tablished. **Read what follows as a boundary theorem and deliberately not a security theorem:** it
225 says exactly how much an invariance argument about composition can prove, which is preservation
226 of d_2 ’s discriminative mechanism at levels (P1)–(P3). It bounds no ASR, and the counterexamples
227 in the box above are counterexamples to reading it as though it did.

228 **Theorem 1** (Bounded reweighting preserves the downstream discriminative mechanism). *Let d_1*
229 *apply per-client scalar weights $w_i \geq 0$ with ratio $\rho := \max_i w_i / \min_{i: w_i > 0} w_i$ (over strictly positive*
230 *weights), operating on n_b benign and n_a adversarial updates in a round of size K . Assume $w_a > 0$*
231 *for the adversarial clients; the $w_a = 0$ case is Lemma 1. Then:*

232 1. **Rank aggregators.** *The composition passes $\{w_i u_i\}$ to an unweighted order statistic over*
233 *K transformed values per coordinate. The mechanism is preserved when:*

- 234 (i) Count majority: $n_a < K/2$.
- 235 (ii) Signed ordering (C2): *on suppression-relevant coordinate k , the adversarial trans-*
236 *formed value lies strictly beyond every benign one: $w_a u_{a,k} > \max_b w_b u_{b,k}$ (upper*
237 *tail) or $w_a u_{a,k} < \min_b w_b u_{b,k}$ (lower tail).*

238 Sufficient magnitude condition for (ii): *if $\text{sign}(u_{a,k})$ is common to all adversaries and*
239 *$|u_{a,k}| > \rho \cdot \max_b |u_{b,k}|$, then (ii) holds. This branch is one of two mechanism-preserving*
240 *regimes, the other (strong attenuation, $w_a \rightarrow 0$) being Lemma 1.*

241 2. **Geometric median (RFA):** *the mechanism is preserved when separation dominates*
242 *displacement—the minimum adversarial–benign separation among the transformed up-*
243 *dates strictly exceeds twice the benign spread plus the displacement the adversaries induce*
244 *in the geometric median (Appendix G defines the four quantities and proves it)—under*
245 *which every adversarial residual exceeds every benign one, so RFA assigns adversarial*
246 *updates strictly lower inverse-residual weight.*

247 *Proofs are in Appendix G. Both parts are sufficient conditions for d_2 to preserve the downstream*
248 *suppression ordering, not to exclude adversarial contributions outright; **neither is sufficient for low***
249 *ASR, and part (2) constrains the aggregate adversarial coefficient only through Corollary 1.*

250 Theorem 1(2) orders residuals *per client*, which alone does not bound the adversary’s *total* share
251 of the aggregate. A realized *separation margin* $m \geq 1$ —the factor by which that condition is met,
252 defined with the bound in Corollary 1 (Appendix G)—does bound it, but only quantitatively: as

253 $m \rightarrow 1$ the bound degrades to the adversarial count fraction, so **the separation condition on its**
 254 **own is not enough.** This is the one place the theory predicts a number and the measurement lands
 255 inside it: at the measured $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds)
 256 the bound is $\Lambda_a \leq 12.6\%$, against a realized $\Lambda_a \leq 1.8\%$. The $m \rightarrow 1$ boundary is where §6’s
 257 heterogeneity failure sits, and it is reached ($m=1.01$ at $\alpha=0.1$).

258 Λ_a is level (P4) in measurable form. (P4) is the first level whose content is the adversary’s in-
 259 fluence on the aggregate, and for a geometric aggregator Λ_a is that level made measurable: Corol-
 260 lary 1 bounds it, Lemma 1 drives it to a benign-only convex combination, and §4’s instruments
 261 intervene on it directly. **So the level the evidence points to is one the theory can bound—just**
 262 **not from a (d_1, d_2) definition pair alone,** which is what a screen would need. Unqualified, “ad-
 263 mission” means *admitted adversarial mass*, whether an adversarial update enters d_2 ’s aggregate
 264 at all; the three aggregator-specific quantities Definition 2 sets out are never used interchangeably
 265 (Appendix E).

266 The remaining case dominates in practice—FoolsGold drives the adversarial weight to zero. This
 267 does *not* remove the adversary from RFA’s input (Definition 1): RFA receives a zero vector, a point
 268 of the geometric-median problem like any other. The correct statement is stronger than removal:

269 **Lemma 1** (Zero-weight annihilation). *Our implementation runs the ε -smoothed Weiszfeld iteration,*
 270 *whose fixed point $\mu' = \sum_i \lambda_i u'_i$, $\lambda_i \propto 1/\max(\|u'_i - \mu'\|, \varepsilon)$, is a convex combination for every*
 271 *configuration, so no non-degeneracy assumption is needed (Appendix G). If $w_a = 0$ for every ad-*
 272 *versarial client then $u'_a = 0$ for each, and writing $\Lambda_a := \sum_{a \text{ adv}} \lambda_a$ the adversarial terms vanish*
 273 *identically:*

$$\mu' = \sum_{b \text{ benign}} \lambda_b u'_b = (1 - \Lambda_a) \bar{u}, \quad \bar{u} := \sum_b \frac{\lambda_b}{1 - \Lambda_a} u'_b,$$

274 *a convex combination of benign transformed updates. The aggregate is a **contraction of a benign-***
 275 ***only weighted average:** u_a does not appear in μ' at all, and the origin point’s only effect is a smaller*
 276 *effective step—a utility cost, not a security failure.*

277 The same argument covers CoordMedian. The two cases are exhaustive over every measured round
 278 (FG zeroes all adversaries in 32% of adversary-present rounds, the separation condition holding in
 279 the other 68%), and the mechanism survives for *opposite* reasons: under Lemma 1 the adversaries
 280 hold most of the Weiszfeld mass ($\Lambda_a = 0.71$) but carry no payload; under Corollary 1 they carry
 281 payload but at most 12.6% of it. The contraction $1 - \Lambda_a = 0.29$ is composition’s accuracy cost:
 282 FG→RFA reaches accuracy 0.50 and FG→CM 0.66 against 0.78 for the aggregators alone, paid
 283 even when the zeroed clients are benign.

284 2.2 Heterogeneity is not the mechanism: ρ against Δ_c

285 **The finding, stated first.** What costs a downstream defense its suppression is not how heteroge-
 286 neous the upstream transform is but **whether its reweighting is correlated with adversary iden-**
 287 **tity**—and not even the size of that correlation, which has **two opposite regimes**. This is why our
 288 own headline predictor failed.

289 **Two different things a coefficient vector does, and the theory sees only one.** (Symbols: Ap-
 290 pendix E.) Proposition 1 turns on **dispersion**, $\rho := \max_i c_i / \min_i c_i$ —how far apart the coefficients
 291 are, which decides whether an ordering survives—and says nothing about *which* clients receive
 292 which coefficient. That second property is **adversary-correlated dispersion**,

$$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A], \quad A \text{ the adversarial participants},$$

293 and the two are **independently controllable**, not merely uncorrelated in our sample: a transform
 294 can spread the coefficients arbitrarily at $\Delta_c = 0$ exactly, or leave them nearly equal while shifting
 295 the adversary’s share, and Proposition 5 exhibits a family of each. **Proposition 1 and Theorem 1**
 296 **constrain ρ ; Lemma 1’s attenuation channel is about Δ_c ,** and §5.2 finds the real transforms differ
 297 on the second. Δ_c is *not* a screening quantity: computing it requires knowing A (§8).

298 **Two mechanism-preserving regimes, at opposite ends of ρ .** Lemma 1 is a second, inde-
 299 pendent route to suppression, and the two together explain all three PASS compositions. Let
 300 $r := w_a / \max_b w_b$ and S be the attack’s coordinate extremeness. **(A) Ordering preservation**

keeps the adversary in the tail where a rank aggregator discards it ($S \cdot r > 1$). **(B) Attenuation** instead drives the adversarial transformed value *inside* the benign spread, where it cannot be an extreme order statistic and its payload is scaled by r ($S \cdot r \ll 1$; $r = 0$ is exact annihilation). The hazard is $S \cdot r \approx 1$, and our two weighting defenses sit at opposite ends: FoolsGold’s weights are bounded ($\rho \leq 2.1$, $S \cdot r \geq 4.8$), regime A; Reputation’s are wildly unbounded (ρ mean 1.4×10^5) precisely *because* it crushes the adversary ($S \cdot r = 0.017$), regime B. So Reputation→CoordMedian violates the ordering condition by five orders of magnitude and still suppresses (Figure 3, App. J, exhibits both regimes). Hence ρ does not order Δ_c (Proposition 5), and Δ_c ’s *magnitude* does not order suppression either, the same large $|\Delta_c|$ being protective at one end and catastrophic at the other. **Neither quantity is a predictor**; they buy an account of *why* a heterogeneity-based reading mispredicts.

3 The Testability Boundary

Before reporting any experiment we have to say which experiments are *admissible*. A screening framework of this shape imposes a structural limit on its own validation, and the limit is characterizable: it *derives* the class of designs in which C2’s contribution is identifiable at all, and every experiment below is a member of that class.

Proposition 2 (Why constituent-effect contrasts cannot identify composition effects under inherited-effect gating). *Let a screen certify ($d_1 \rightarrow d_2$) against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses’ definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P ’s contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and **the cell needed to separate P from d_2 ’s standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P ’s contribution. Identification requires holding d_2, a and therefore E fixed while intervening on the upstream transform: **the admissible designs are within-defense**.***

The structure is a collider: the gate E is a function of constituent effectiveness, effectiveness affects the composition’s outcome, and the mechanism condition P is fixed by the same pair of definitions—so conditioning on E makes P and effectiveness co-vary across every certified candidate. The design table is the whole argument, and **the missing counterfactual is a cell, not a sample size**:

	$M(d_2, a) < \theta$ (d_2 suppresses a alone)	$M(d_2, a) \geq \theta$ (it does not)
P holds (mechanism preserved)	observed: every certified pair	empty by the gate
$\neg P$ (mechanism destroyed)	observed: predicted-FAIL pairs	empty by the gate

Reading down a column contrasts P against $\neg P$ at fixed effectiveness—the comparison that would identify P ’s contribution—and the right-hand column, the only one in which d_2 is not already doing the work, is empty for every menu $\mathcal{D}$. **The counterfactual needed is excluded by design, not undersampled.** The emptiness follows from the gate, not the menu, and nothing in the argument is specific to FL, to backdoors, or to our conditions (Appendix G). Its instance for *our* criterion is one line:

Lemma 2 (C0 and C1 place the effective constituent downstream). *Let ($d_1 \rightarrow d_2$) satisfy C0 and C1, and let a^* be a committed attack witnessing C0, so $\text{ASR}(d_1, a^*) \geq 0.5$. Then $\text{ASR}(d_2, a^*) < 0.5$: the downstream defense suppresses a^* on its own.*

Instantiated: in *every* in-scope pair our conditions certify, d_2 already suppresses the attack d_1 fails, so a certified composition’s low ASR is always attributable to d_2 ’s standalone strength (C1) rather than to d_1 having preserved d_2 ’s mechanism (C2, C3)—confounded **by the conditions themselves**, not by an unlucky menu. We name this **mechanism–effectiveness confounding**.

This is selection on the outcome; where it hides is the point. The hazard is standard—conditioning on a descendant of both the putative cause and the outcome [Elwert and Winship,

2014, Pearl, 2009]—and **we claim no novelty for it**. What is not standard is that in composition studies the gate is presented not as conditioning but as a *reasonableness* filter—“we study defenses that work”—so it reads as hygiene rather than as the step that removes identification. Two consequences are routinely assumed away: **a better estimator cannot fix it**, no reweighting of occupied cells recovering an unoccupied one, and **the gate cannot be dropped**, because without it C1 is vacuous.

Both worked instances are ours, and one of them inverted a sign. We name no other paper; our own two designs are the instances. The C1-gated study of §5 is the template at the level of the *criterion*, and the dose ladder of §5.2 is a second instance one level down, where the gated quantity is the dose’s incidental attenuation of whichever client carries the poison. **We ran the outcome-gated design first, and on the identical cell it returned the opposite sign** (ladder -0.272 against instrument $+0.098$, Table 2): the hazard cost us the sign of our own effect, in the direction that flattered us.

The admissible design, derived—and this, not the criterion, is the practical contribution. Read forwards the boundary is *constructive*: each clause below is forced by a term in Proposition 2, not chosen for convenience. **(1) Hold d_2 and the attack fixed**, so the gate’s input $M(d_2, a)$ is numerically the same quantity on both sides of every contrast and cannot co-vary with the mechanism. **(2) Intervene on the upstream transform** instead of selecting a different d_2 off a menu, because it is varying d_2 that empties the identifying cell. **(3) Pin the adversarial coefficient channel**, because an upstream that attenuates the adversary moves the outcome by a route with nothing to do with d_2 ’s statistic—the confound that inverted our own sign above. **(4) Measure every channel—aggregate displacement, decision, admission, influence—rather than the outcome-conditional contrast**, the outcome being exactly the variable the gate conditioned on. §4 is clause (3) built as an instrument; §8 states all four as a numbered procedure for any composition claim.

A second boundary: what such a screen cannot find. The gate that limits what a screen can be *validated* on also limits what it can *cover*—which converts the most obvious objection to our criterion, that its recall is poor, into a scope result.

Proposition 3 (Inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$ ’s suppression of a **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Proposition 2. Hence, writing S for the compositions with $M < \theta$ and $\mathcal{E} \subseteq S$ for the emergent ones, any such screen’s recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|S|$ on every menu and every attack.*

This is a scope theorem, not a defect found by tuning, and it is specific neither to C1 nor to FL: any screen requiring a constituent to suppress the attack independently inherits it. Our own lowest-ASR composition is the witness—FG→RFA reaches 0.093 where FoolsGold alone reaches 0.732 and RFA alone 0.845—so the bound binds at $|\mathcal{E}|/|S| = 1/5$ (§6). Screening for inherited robustness and searching for synergistic compositions are different problems; only the first is ours.

What the boundary rules out, our own previously-decisive test included. These proofs are short and their value is what they rule in and out. They rule *out* the experiment we had previously named as decisive—a d_2 whose signal is invariant under positive rescaling but *weak in isolation* cannot satisfy C1 in scope, so no such certified pair exists to test. §4–§5 run four members of the class they rule in. **The strongest disturbs the statistic while pinning the adversarial mass, isolating statistic disturbance from adversarial coefficient attenuation—so we built it (§4).**

4 Statistic Preservation Is Not Security Preservation

The arc that produced this section. We first froze a pooled, statistic-level hypothesis (H1): across a within-defense ladder, ASR should rise with measured disturbance of the statistic the downstream defense reads. A pre-registered ladder **rejected it**, the effect coming out *negative*— -0.272 , $p_{\downarrow}=0.0015$ on the flagship cell, ASR falling as the dose rose in three arms of four—because it moved adversarial influence and the statistic *together* and so never tested H1 at all (§5.2). This section builds the instrument that fixes it, pinning the adversarial coefficient share while the statistic moves, so statistic disturbance is isolated from adversarial coefficient attenuation. C2 is written

at level (P2); if losing (P3) meant losing (P5) the criterion would transfer straight to security, and the experiment below adjudicates that implication. **What it establishes it establishes by oracle intervention on two arms in two invariance classes** (*Causal* tier)—not a law about defense compositions in general.

An instrument that closes the attenuation channel by construction. A positive rescaling reaches d_2 through two channels at once: it disturbs the *statistic*, and it *attenuates* whichever client carries the poison (Lemma 1). Separating them needs an upstream that moves the first while pinning the second. **Mode S** does this: it fixes every adversary’s coefficient at $c=1.0$ exactly and spreads only the benign clients over $\rho = e^{2\kappa}$, so the **adversarial coefficient share is constant at 0.266667 across all four rungs** (spread 3.5×10^{-7}), and **a rise in ASR here cannot be attenuation**. Design and thresholds were frozen before any run, the share verified with no ASR involved (Appendix S). **Mode A** sweeps the payload instead (§5.3). **Both are oracle instruments, not defenses and not models:** they read adversary identity, which no deployable defense can, and they do **not** quantitatively reproduce any real defense’s behaviour—at matched disturbance two real upstream anchors sit $+0.767$ and $+0.789$ above the synthetic curve (§5.2), the measured size of that gap. Their purpose is causal identification, not defense evaluation.

Two readings of C2, frozen against each other. *H-statistic:* what matters is disturbance of d_2 ’s statistic, read off its *decisions*. *H-admission:* what matters is the *adversarial input* d_2 *admits*. On the Krum arm these predict opposite outcomes, since Mode S disturbs Krum’s selection heavily while—the adversary’s weight being pinned—leaving the adversarial mass it admits untouched. That arm adjudicates (Table 3, App. J): as ρ climbs to 54.6, Krum’s **decision changes in up to 0.80 of rounds** while its **admission changes in exactly 0.000**, and **ASR shows no evidence of the predicted increase, remaining inside the pre-specified equivalence margin** ($p_{\uparrow}=0.70$; $|\Delta|=0.026 < 0.15$). **H-statistic is refuted:** on this arm **substantial statistic and decision disturbance does not produce a corresponding change in suppression once the adversarial coefficient share is held fixed, and admission is the channel the evidence points to.**

What the instrument does *not* hold fixed, named in full—and what it pins exactly. Mode S pins the *adversarial* coefficients; it does not pin the benign ones and was never claimed to. So the dose also moves (a) Krum’s pairwise distances, hence the statistic itself; (b) *which* benign client Krum selects, in 0.533/0.800/0.733 of rounds at $\rho=2.72/7.39/54.6$; (c) the **magnitude and geometry of the update actually aggregated**, since the selected client enters scaled by its own coefficient; and (d) the model trajectory, which diverges across rungs in consequence. Channel (c) is **already measured and printed:** the Δ agg. column of Appendix B is $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, the relative displacement of the aggregate the defense *emits*, and reads **0.892** here. Appendix C splits that 0.892, as vectors, into 0.986 of *re-selection* (a different benign client is chosen, and in 0/11 of the rounds where that happens it is an adversary) and 0.973 of *rescaling* (the chosen client arrives scaled by its own coefficient); for *cos_krum* the displacement is *entirely* rescaling, since 0/15 selections change at any rung. The claim this arm licenses is therefore sharper: **a 0.892 displacement of the emitted aggregate, together with decision changes in up to 80% of rounds, leaves suppression inside the pre-specified equivalence margin—provided the adversarial coefficient share does not move. What is pinned is pinned exactly:** $c_{\text{adv}}=1.0$ at every rung with maximum deviation 0.00e+00 and the share constant to $3.5\text{e}-07$ on CIFAR-10, $3.9\text{e}-07$ on FEMNIST—float32 read-back noise, not slack in the instrument (`measure_admission.py`).

A pre-registered replication, on a different defense class, that refutes the positive half. The two existing Mode-S arms move together on both predictors, so neither adjudicates. `coord_median` under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the **smallest decision change** (0.482) and the **largest admission change** (0.033), so the two readings predict opposite rank positions for its rise. Thresholds were frozen against the two published rises—confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$ (Appendix S). **The outcome is $\Delta = +0.098$, so the admission ordering is refuted:** the arm with the largest admission change rose *less* than one with a smaller change. Nor does the decision reading predict it, having called the *smallest* rise when $+0.098$ is the middle one; the added cells leave the pooled model unconfirmed ($R^2=0.239$, the share coefficient’s sign failing the pre-specified conjunction). **What replicates is the negative,** on a coordinate-wise order statistic rather than Krum’s selector: a decision change of 0.482 costs no suppression by the frozen margin, at accuracy 0.698. Two qualifications: this cell’s identity ASR is 0.443, so

equivalence is to the *identity rung*, not a low absolute ASR; and the rise is positive in 5/5 seeds (post-hoc), so the cost is **small and consistent, not zero**. **The instrument earns its keep**: the confounded ladder on this identical cell *falls* 0.272 ($p_{\downarrow}=0.0015$), so closing the attenuation channel **inverts the sign of the effect**.

What the two arms jointly establish. Both halves are witnessed by runs already reported, so the following formalizes measurements rather than adding a claim.

Proposition 4 (Statistic-level non-identifiability). *Within the positive per-client rescaling class there exist transform families T_1, T_2 and downstream defenses d_2 such that (a) T_1 and T_2 induce arbitrarily different statistics and decisions at d_2 while inducing the same admitted adversarial mass and the same suppression; and (b) T_1 and T_2 induce the same admitted adversarial mass—indeed the same statistic and the same decision—while inducing different suppression. Consequently no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression.*

Both parts are proved by exhibited witnesses (Appendix G): Mode S into Krum under model-scaling for (a), `cos_krum` for (b). **The corollary is the paper’s title claim**: no metric defined *solely* on preservation of the downstream statistic can identify security equivalence, (a) exhibiting statistic disturbance without security change and (b) the converse. **Scope**: a non-existence claim *within the positive-rescaling class on these defenses and attacks*, by exhibited counterexample—**not** that no metric of any kind can order security.

Proposition 5 (ρ and Δ_c are independently controllable). *Within the same class, the statistic-level dispersion $\rho = \max_i c_i / \min_i c_i$ and the adversary-correlated dispersion $\Delta_c = \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ are not functionally related: there is a family with $\Delta_c \equiv 0$ exactly while ρ ranges over $[1, 54.6]$, and a family with the statistic’s dispersion held fixed while Δ_c ranges over $[-1.138, +2.469]$. Hence neither quantity determines the other.*

This makes §2’s warning precise: **a transform reported by ρ alone leaves Δ_c entirely free**. It does *not* follow that Δ_c is a predictor—its magnitude fails to order suppression either (§5.2). The instrument is elaborate because §3 leaves nothing simpler. §5 answers what remains: *what happened when we tried it the obvious way first?*

5 What the Boundary Licenses, and What We Found

We validate on CIFAR-10, $N=10$ clients, $K=5$ per round, $f=0.2$, 50 rounds, 4-layer CNN, Dirichlet $\alpha=0.5$ (non-IID), two committed attacks (model-scaling, pixel backdoor); full setup in Appendix S. **Convention**: unless stated otherwise ASR means *max-committed ASR*, the maximum over the two attacks of the per-attack seed mean, thresholded at 0.5, so “suppresses” means the backdoor is not the model’s dominant response. Sample sizes, the accuracy floor (a destroyed model has a low ASR for the wrong reason), the equivalence-margin rule and the meaning of “refuted” are fixed in Table 1. Inside the *Screening* tier the 42 pairs split four ways, of very unequal weight and never pooled (Table 6, App. K): 18 *development* pairs, so **fit, not prediction** (16/18); 24 pre-registered held-out pairs bearing on specificity only (22/24); 7/7 out-of-sample PASS predictions with *no* informative negative controls; and §5.1’s prospective suite, **the only tier with both informative negatives and a real risk of falsification** (13/20).

5.1 Two attempts that the confound defeated

Both were built to validate the *screen*; neither could, so they are evidence for Proposition 2 rather than for the framework.

Two pre-registered suites, and the two ways the boundary asserted itself. (i) *A prospective suite on three unused defenses*: 13/20, and *no positive test*. Everything scored in §6 was developed on the menu it is scored on, so we ran a prospective test on FLTrust [Cao et al., 2021], Krum and Multi-Krum [Blanchard et al., 2017], for which the theory predicts *opposite* outcomes. All 20 labels were git-committed before the output directory existed and both attacks clear the FedAvg sanity gate, so the negatives are informative (11/13 in scope, 5/7 on the falsifiable arm; Appendix M). **Two disclosures dominate it**: FLTrust alone suppresses both attacks (0.048), so its 7 pairs are **out of**

scope by C0—applied *retrospectively*—and its 6 in-scope predicted-LOW pairs are **near-identities**, four bit-identical to FLTrust alone, so **the suite supplies no falsifiable prospective positive test**. (ii) *A matched metric-swap ablation, whose premise its own baselines refuted*. To break the confound by construction we held the suppression *principle* fixed and varied only the invariance class, mirroring Reputation and Krum scoring from *directions* alone. Measuring the C1 input before freezing **refuted the design’s premise**: `cos_reputation` collapses ($0.017 \rightarrow 0.982$) and `cos_krum` reaches 0.078 only at 0.146 accuracy, failing the C1 gate applied everywhere else—**magnitude information is what lets this family handle a magnitude attack**—so C1 bound for all eight pairs, every label was HIGH, and the suite bears on specificity only (8/8; App. H, L). **The confound re-materialized inside the experiment built to remove it.**

The one contrast that escapes—and the design it dictates. One comparison evades the boundary: hold d_2 and the attack fixed and vary only the upstream, so C1 is identical by construction. On *Krum*/model-scaling, NormClip disturbs its selection 0/9 and reaches 0.062 while RFA disturbs it 6/9 and reaches 0.779 (Welch $p=0.012$, $n=5$), with no overlap across the six cells where standalone d_2 genuinely suppresses. **This is post-hoc** (Appendix H) but has what nothing above it does: C1 held exactly fixed, disturbance measured on the mechanism rather than read off the outcome.

5.2 H1, and the confounded design that motivated the instrument

Both suites above tried to validate C2 by *choosing a different downstream defense*—the move Proposition 2 rules out. The admissible alternative varies only the *dose* of upstream disturbance at fixed d_2 and attack, so the C1 input is numerically the same quantity at every rung and the confound is **impossible by construction** rather than matched away.

The instrument, the abscissa, and the predictions frozen against them. The synthetic d_1 has one parameter, the *dispersion* of a positive per-client rescaling: $c_j \propto \exp(\kappa(2j/(K-1) - 1))$ at position j of a (seed, round)-keyed permutation, normalized so $\text{mean}_j(c_j) = 1$. It is Proposition 1’s form, the **dose is** $\rho = e^{2\kappa}$, $\kappa=0$ is bit-identical to d_2 standalone, and normalizing the mean while keying the permutation on (seed, round) holds aggregate scale fixed and stops the dose parking adversaries at a fixed end—neither of which, as it turns out, pins Δ_c . Frozen grid $\kappa \in \{0, 0.5, 1.0, 2.0\}$, four cells across three invariance classes. What matters to C2 is not ρ but what the rescaling does to the statistic d_2 reads, so we measured that first, with no ASR involved, establishing that *there is a dose* (0 at identity, rising to 0.87/0.93 for class (c)) and that *class-(a) invariance holds in code*. *Primary, pooled*: across all 16 (arm, κ) cells the ASR rise over that arm’s own $\kappa=0$ rung is positively rank-correlated with measured disturbance (Spearman, one-sided, $\alpha=0.05$). *Secondary, per arm*: monotone rise for class (c), **FLAT**—the frozen label, scored as equivalence—for class (a), shallower for class (b); one-sided Jonckheere–Terpstra at $n=5$, “flat” scored as *equivalence* ($|\Delta| < 0.15$, every rung below 0.5), a margin fixed in advance. **The ladder was falsifiable both ways**: `cos_krum`’s aggregate is still $c_j u_j$ for the selected j , so suppression can be lost through the magnitude channel, and a class-(c) arm inside the margin would refute the central fragility claim. **No frozen prediction uses S** : measured, S is **not a stable per-attack constant** (medians 0.57/0.92, range to 12.59), a limit on the theorem’s checkability. Design, grid, arms, shapes and refutation criteria were frozen in advance (Appendix I).

Outcome: one arm confirmed, two refuted, one indeterminate—the pooled test fails. All 80 runs completed and were scored per seed against the frozen rules (Table 4, App. J). The **primary** pooled prediction is **refuted**: Spearman $\rho_s = +0.351$ over 16 cells, one-sided $p = 0.091$, against a frozen $\rho_s > 0$ at $p < 0.05$; **in three of four arms ASR falls as the dose rises**. Arm 2 confirms decisively—reputation $0.017 \rightarrow 0.779$ ($p = 1.7 \times 10^{-6}$) at nearly constant accuracy, the paper’s only prospective positive for C2, and it establishes the *destruction* half only. Arm 1 refutes more than its own shape: Krum, whose statistic the fragility claim is stated over, shows no rise—inside the equivalence margin—despite measured disturbance 0.867, and does not reproduce our within-defense contrast ($0.062 \rightarrow 0.779$ under RFA at a *lower* disturbance), so selection-flip rate cannot be its mechanism. Arm 3 is indeterminate and uninterpretable anyway (`cos_krum/pixel` falls $0.490 \rightarrow 0.086$, the last rung below the accuracy gate) and arm 4 is refuted outright (`coord_median/pixel` falling $0.443 \rightarrow 0.170$ at accuracy ≥ 0.69). Both fell where a rise was frozen, which is why neither is scored in our favour.

558 **Two faults, both ours—and now measured rather than conjectured.** *The instrument.* Normalizing $\text{mean}(c)=1$ fixes the aggregate’s scale but not any client’s *relative* weight, and because the
559 dose ignores adversary status a $55\times$ spread mostly *dilutes* whichever client carries the poison: on
560 the same rounds the ladder’s Δ_c drifts monotonically negative with the dose ($0.000 \rightarrow -0.140 \rightarrow$
561 $-0.240 \rightarrow -0.336$), so three arms measure attenuation rather than C2 disturbance. **The adversary-**
562 **targeted dose is the fix:** Mode S is the Δ_c -**pinned** instrument (Δ_c exactly 0 at every rung, ρ up to
563 54.6) and Mode A the Δ_c -**swept** one. *The abscissa.* At matched disturbance two of six *real* upstream
564 anchors miss the synthetic curve badly (RFA $\rightarrow$ Krum $+0.767$, NormClip $\rightarrow$ Reputation $+0.789$), and
565 not through a range mismatch; Δ_c locates the discrepancy, as Proposition 5 says it may, those two
566 being the most extremely targeted of the four real transforms ($-1.21, -0.72$) at ρ of only 14.8 and
567 4.19 against the ladder’s 54.6—**matched on the statistic, badly unmatched on the targeting. But**
568 **this rescues no predictor:** Δ_c ’s magnitude does not order suppression either, and FoolsGold’s is
569 *positive*—**post-hoc and exploratory**, computed with no ASR and rescoring nothing (Appendices J,
570 N). Two checks did pass: the $\kappa=0$ rung reproduces every standalone baseline exactly per seed, and
571 cos_krum ’s selection never moved (0/120).
572

573 5.3 The rest of the targeted suite: the payload dose, and what it does to the ladder’s one 574 confirmation

575 **What Mode S does to the ladder’s one confirmation.** Mode S shows most of §5.2’s one confir-
576 mation was attenuation. With the adversary’s share held fixed, reputation’s rise at $\rho=54.6$ shrinks
577 from the ladder’s $+0.762$ to $+0.178$ (one-sided Welch $t = -5.22$, $p=0.0017$ that it is smaller),
578 and what remains tracks a small *admission* change ($0.000 \rightarrow 0.020$) rather than the large decision
579 change ($0.000 \rightarrow 1.000$), so H-admission is confirmed here too (Table 3). The class-(a) control
580 cuts the other way and refutes us: cos_krum ’s selection is bit-identical at every rung (decision and
581 admission both 0.000), yet ASR is 0.584 at the identity—it never suppresses—and *falls* 0.425 across
582 the ladder ($p_\downarrow=0.0008$; the $\kappa=2$ rung is below the accuracy gate). Exact invariance of the statistic
583 neither implies suppression nor holds it fixed; the magnitude channel moves it regardless, refuting
584 Proposition 1’s practical content without touching its algebra.

585 **Mode A: the two regimes, and where they hold.** **Mode A (payload-only)** closes the other chan-
586 nel: it keeps the benign weights uniform and sets the adversary-to-benign weight ratio to $r = e^\nu$
587 at fixed mean, so the adversarial share sweeps $0.049 \rightarrow 0.710$ with the statistic’s dispersion held
588 fixed—the mirror of Mode S. That gives Theorem 1 a single falsifiable shape: ASR single-peaked
589 at $r=1$, since an over-attenuated adversary carries no payload (Regime B) and an over-amplified
590 one is discarded by a rank aggregator (Regime A). It is **confirmed for** coord_median , the rank
591 aggregator the regimes are stated for: ASR rises $0.124 \rightarrow 0.315 \rightarrow 0.443$ into $r=1$ and falls
592 $0.443 \rightarrow 0.296 \rightarrow 0.098$ past it, both sides significant ($p_\downarrow=3.0\times 10^{-4}$, 6.4×10^{-4}) at accuracy
593 ≥ 0.74 (Figure 3). It is **refuted for reputation**, which is monotone—maximal at the most attenu-
594 ated rung (0.894 at $r=0.135$)—because an attenuated adversary reads as *consistent* to a reputation
595 score. Reputation is not a rank aggregator, so the shape was never licensed for it.

596 **The same channels, across four aggregators of three structural kinds.** Across selectors on pair-
597 wise and on cosine distance (krum , cos_krum), a coordinate-wise order statistic (coord_median)
598 and a weighted averager with cross-round state (reputation), the negative is **not Krum-specific**: the
599 decision channel moves by 0.482–1.000 on three of the four while **admitted adversarial mass stays**
600 **at exactly 0.000 on all four**, and the only two arms whose ASR *rises* are the only two with nonzero
601 $\Delta\Lambda_a$. **An analysis of runs already reported, not a new experiment, and it orders nothing:** the
602 larger $\Delta\Lambda_a$ (0.033) belongs to the arm with the *smaller* rise (Appendix B).

603 **A second dataset and architecture, pre-registered.** The same cell on EMNIST-
604 byclass/simple.cnn in place of CIFAR-10/cifar.cnn, frozen in git before any FEMNIST
605 ASR existed. Premise measured first, no ASR anywhere: the decision moves on 0.643–0.714 of
606 rounds, admitted mass exactly 0.000 at every rung. **Suppression does not move**— $\Delta = -0.017$
607 against the frozen ± 0.15 margin, beside CIFAR-10’s -0.026 , at accuracy $0.767 \rightarrow 0.835$, so
608 the gate never binds. **Replicated on a second dataset and architecture**—the frozen wording,
609 never “generalizes”. Two limits from the freeze: the Jonckheere–Terpstra was specified to catch a
610 monotone *rise* and the trend is a *fall* ($z = -1.563$), so it is not read as support for flatness; and with

611 the identity rung at 0.027 the refuting branch was reachable (> 0.177) while the indeterminate one
612 was not, so equivalence is to the identity rung, not a low absolute ASR.
613 It addresses neither recall nor discovery, and nothing in §5.2 is rescored.

614 6 What the Causal Boundary Permits—and What It Does Not

615 **This section is a consequence of the boundary, and what survives is an *evaluation-prioritization***
616 ***criterion*—not a security screen.** $C1 \wedge C2$ answers one question: given a menu, which ordered com-
617 positions are worth running first. **We disclaim predictive validity explicitly**—it does not predict
618 whether a composition is secure, its positive direction is unvalidated, and the cross-arm admission
619 ordering a predictive reading would need was *refuted* by our own pre-registered replication (§4). It
620 also costs us our own best composition: under the per-attack quantifier $FG \rightarrow RFA$ **no longer satis-**
621 **fies C1**, so the lowest-ASR composition we found is a **false negative** and Proposition 3’s witness—
622 uncertifiable by *any* inherited-suppression screen rather than missed by ours—while the certified in-
623 distribution set ($FG \rightarrow CM$ 0.107, $Rep \rightarrow CM$ 0.423) is correct and the miss costs nothing detectable
624 (0.093 against 0.107, Welch $p=0.52$). Out of sample it licenses *positive* predictions only, against un-
625 informative FAIL controls, so **this is not an out-of-sample classification result**: the *Screening*-tier
626 38/42 at recall 2/5 rests on the in-distribution tiers. The defensible claim is *conservative prioritiza-*
627 *tion*; scoring condition by condition, the confusion matrix, the false negatives and what the rule can
628 and cannot do are in Appendices O–P, Q and Table 8 (App. R).

629 **The procedure, and where it breaks.** (0) If one defense already suppresses every attack you
630 have, stop (C0); (1) require a constituent that suppresses the committed attack (C1); (2) read C2 off
631 the downstream signal’s *invariance class*, which §3 says you cannot test directly; (3) read C3 as a
632 margin from a short pilot, never as a gate. Then **evaluate the survivors first—and evaluate them**:
633 none of this certifies security. **Where the rule breaks is measured rather than argued: under**
634 **extreme heterogeneity (Dirichlet $\alpha=0.1$) two of three certified compositions cross the threshold**
635 **and RFA’s all-round mean separation margin collapses from $m=3.24$ to $m=1.01$ —the $m \rightarrow 1$**
636 **limit at which Corollary 1’s bound says nothing.** Theorem 1’s guarantee is therefore not merely
637 conservative under heterogeneity but *empty* there, and the theorem earns its keep by predicting its
638 own failure region: a short pilot measuring $m \approx 1$ declines to deploy. **Post-hoc, not pre-registered**
639 (Appendix K.1).

640 7 Related Work

641 **The contrast in one sentence.** Prior work asks whether a defense withstands an attack. We ask
642 whether an *evaluation criterion* can causally attribute the security of a composition to preservation
643 of the downstream mechanism.

644 **Byzantine-robust aggregation, studied in isolation.** Krum [Blanchard et al., 2017], Trimmed
645 Mean [Yin et al., 2018], RFA [Pillutla et al., 2022], FoolsGold [Fung et al., 2020], FLTrust [Cao
646 et al., 2021] and FLAME [Nguyen et al., 2022] are each evaluated as a single stage; we propose no
647 new defense and reason about *ordered pairs* of existing ones. Ensemble adversarial training [Tramèr
648 et al., 2018] *assumes* upstream processing preserves downstream signals; we make that assumption
649 explicit, checkable, and—as it turns out—**the wrong abstraction for security**.

650 **Adaptive attacks are why an inherited-robustness guarantee is not a security guarantee.** Op-
651 timized model poisoning defeats Krum, trimmed mean and FoolsGold once the attack is chosen
652 *against* the defense [Shejwalkar and Houmansadr, 2021, Bhagoji et al., 2019], as do distributed
653 and durable backdoors [Xie et al., 2020, Wang et al., 2020, Zhang et al., 2022]. C1 quantifies
654 over a *committed* attack, so this literature bounds what any inherited-suppression criterion—ours
655 included—can promise; our single whitebox stress test (§8) measures that gap rather than defending
656 the criterion.

657 **Evaluation methodology, which is where this paper sits.** The adjacent precedent is adversarial
658 robustness, where reported numbers proved unreliable until evaluations were required to be adap-
659 tive [Croce et al., 2021]. Our point is methodological in the same way but about **identification** rather
660 than attack strength: composition studies gate candidates on standalone effectiveness [Kairouz et al.,
661 2021] and then read the composition’s outcome as evidence about the mechanism, which Proposi-

tion 2 shows cannot work at *any* sample size. What is new is the level hierarchy at which the preservation intuition fails and the design class that can test it (Appendix D).

8 Conclusion

Downstream-statistic preservation is causally insufficient for attack-suppression preservation, and the screening rule is a consequence of that, not the result. With the downstream defense, the attack and standalone effectiveness held fixed, and the intervention applied *directly* to the statistic that defense reads, changing that statistic can have little or no effect on suppression—and in the same cell the effect **reverses sign** once adversarial influence co-varies with it (-0.272 against $+0.098$). Positive per-client rescaling therefore admits invariance guarantees (Prop. 1) sufficient for preserving the *downstream discriminative mechanism*, never for low ASR; and *any* criterion gated on the outcome it predicts carries a **structural identification boundary** (Prop. 2, Lem. 2 its FL instance) no amount of data removes—**general, where our empirical claims are not.**

What is proved, what is supported, and what is not. **Proved:** the invariance classes (Prop. 1); sufficient conditions for mechanism preservation on the weighting→aggregator class (Thm. 1, Lem. 1); the identification boundary and the design class it leaves admissible (Prop. 2, Lem. 2); the recall bound on inherited-suppression screens (Prop. 3). **Empirically supported** (*Causal*, then *Replication*): that substantial disturbance of the downstream statistic need not produce a corresponding change in suppression when the adversarial coefficient share is held fixed—on the Krum arm under Mode S and, by pre-registered replication, in a *different* invariance class and on a *second dataset and architecture*; and that the criterion behaves as a conservative prioritizer on 42 pairs. **Not established:** its positive predictive direction ($R^2=0.239$, cross-arm ordering refuted), and generalization beyond the scope Table 1 fixes, $\gamma \approx 0.66$ being architecture-conditional. The criterion **trades discovery for reliability**: it rejects working compositions (recall 40%, one of them ours), cannot in principle do otherwise (Prop. 3), and draws negative-direction evidence only from the in-distribution tiers. FG→CM is an evaluation *ordering*, **not a deployment recommendation** (Appendix F).

What to do instead: six steps for any composition claim. (1) Name the downstream mechanism and its invariance class. (2) State *which* preservation level you claim, (P1)–(P5). (3) Check the upstream transform lies in the class your argument covers—C0 here; one altering participation, carrying cross-round state or mixing clients is covered by no invariance argument. (4) Hold d_2 , the attack, and therefore the gate’s input fixed; **do not select candidates on the outcome.** (5) Intervene on the upstream transform and **measure the channel**—aggregate displacement, decision, admission, influence—not the outcome-conditional contrast. (6) Report the level at which the chain broke. Steps (4)–(5) are what Proposition 2 licenses, and following them turned our -0.272 into $+0.098$; **this, not the criterion, is what we ask readers to carry away.** One boundary it does not cross: under *one* whitebox composition-aware *adaptive* stress test FG→RFA rises to ASR = 0.167 ± 0.069 , $3.3\times$ its seed-matched committed 0.051, so **an inherited-robustness guarantee is not adaptive robustness**—the condition the framework most lacks (Appendix K).

The negative is the result. The chain holds only down to (P3) and below that breaks in *both* directions, each with a measured witness. Our own statistic-level hypothesis is refuted pooled, and the admission reading that replaced it survives as a *direction*, not a predictor. **So the object worth screening is not whether an upstream transform preserves a downstream statistic. It is level (P4), the adversarial contribution that survives the downstream defense.** Mode S *measures* (P4), but only by reading adversary identity, which no deployable defense can—as with Δ_c —so our experiments identify the causal channel and **are not a (P4) screen.** That leaves a research question: **can (P4) be estimated without knowing which clients are malicious?** Corollary 1 bounds it from the theory side, so it is open rather than hopeless.

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Table 1: **Scope of claims.** The one place each scope condition is stated. **Claims do not transfer upward:** the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them.

kind	content	status, and what it does <i>not</i> license
General	An evaluation gated on the outcome it predicts cannot identify the mechanism’s contribution; admissible designs are within-defense (Prop. 2, Prop. 3).	Proved, and not specific to FL, to backdoors, or to our conditions. Licenses no claim that a composition <i>is</i> secure, nor that mechanism preservation implies suppression.
Formal	Mechanism preservation as <i>invariance</i> under positive per-client rescalings $T(u_i) = c_i u_i$, $c_i > 0$, into specific rank and geometric aggregators (Thm. 1, Lem. 1, Cor. 1).	Proved on that class only —a <i>laboratory</i> for composition, not a model of FL defenses in general. A d_1 that alters participation, carries cross-round state, mixes clients, or aggregates first has no such T and is covered by nothing here (C0, applied retrospectively to arms frozen before it). Bounds no ASR. and Regime A’s condition is not operationally checkable (S is not a stable per-attack constant).
Empirical	§4’s interventions and §5’s 42-pair menu: CIFAR-10, CifarCNN/ResNet18, $N \in \{10, 100\}$, $f \in [0.05, 0.6]$, Dirichlet $\alpha=0.5$, star topology , three attack families; one arm on FEM-NIST/simple_cnn.	Measured at $n=5$ seeds per cell ($n=8$ for <code>cos_krum</code> , $n=3$ on FEMNIST), fixed before the runs, so intervals are wide and no null is read as flatness without a pre-registered equivalence margin. Dataset transfer is tested on the flagship arm alone—“replicated on a second dataset and architecture,” never “generalizes”. “ Refuted ” always means a pre-registered <i>directional</i> hypothesis was rejected against a rule frozen for it, never that a mathematical claim was disproved.
Both instruments are oracles	Mode S and Mode A build c_i from adversary identity, as does Δ_c .	They identify a causal channel and are not screens ; computing (P4) without adversary identity is our open problem, and admission is a <i>direction to measure</i> , not a validated predictor. C1 and all built on it assume a committed adversary: what is bounded is <i>inherited</i> robustness under a fixed attack, never adaptive robustness .

A The signal-preservation conditions in full

Definition 3 in the body states the four conditions and every disclosure attached to them; what follows is the same criterion with its *stage structure*, which is worth separating because conflating the stages is the main source of confusion about what the framework claims.

Composition ($d_1 \rightarrow d_2$) suppresses attack a when the conditions below hold. The criterion is a **two-stage screen**, and we label the stages because conflating them is the main source of confusion about what the framework claims: stage 1 is empirical screening of the *constituents*, stage 2 is mechanistic screening of the *composition*.

Stage 0 — applicability (C0). Whether the framework has anything to say about this pair at all. This is a condition on the criterion’s *domain*, and Lemma 2 is stated over it together with C1.

(C0) The downstream mechanism is load-bearing. $\text{ASR}(d_1, a) \geq 0.5$ for at least one committed attack a . C2 and C3 concern preservation of d_2 ’s mechanism, so they carry information only when that mechanism does work the composition needs: if d_1 alone suppresses everything, the composition inherits that suppression and a correct prediction is correct for a reason the framework did not supply. The asymmetry is deliberate—a strong d_2 is the intended case. **Failing C0 removes a pair from the criterion’s domain; it never certifies one.** *Provenance:* C0 was vacuous on the menu the framework was developed on (strongest single defense 0.519) and was formulated only after the prospective suite of §5.1 made it bind, so it is applied **retrospectively** and is **not part of what the 42 pairs validate** (Appendix M).

Stage 1 — empirical constituent screening (C1). Read off standalone evaluations. This screens; it does not explain, and it is not a mechanistic claim.

(C1) Individual effectiveness. For *every* committed attack a in the evaluated set, $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a) < 0.5$. The quantifier matters: our headline metric is the *maximum* over attacks, so requiring a suppressing constituent for only one attack would license compositions that fail on the other. C1 is defined through the same ASR threshold that constitutes the outcome, so it is an **empirical prerequisite, not a mechanistic property**—the one condition that cannot be checked without running something.

787 **Stage 2 — mechanistic composition screening** (C2, with C3 as a diagnostic). Determined by d_1 's
788 and d_2 's definitions alone, no experiment required, and what Theorem 1 proves. **The evaluation-**
789 **prioritization criterion is C1∧C2**, with C3 retained as a **margin diagnostic** rather than a gate for
790 the reasons scored in Appendix Q. *The pre-registered labels of §5 were frozen with C3 in the gate*
791 *and are not relabelled*, and the demotion changes no reported number.

792 **(C2) Discriminative-property preservation.** d_1 's transform does not destroy d_2 's discriminative
793 property, as determined by Proposition 1: automatic for cosine-similarity downstream de-
794 fences, conditional for coordinate ordering, and violated for consensus distance under het-
795 erogeneous rescaling.

796 **(C3) Margin preservation.** d_1 does not compress the quantitative margin below d_2 's operating
797 threshold. For rank aggregators: extremeness gap survives reweighting. For geometric
798 aggregators: separation condition holds on transformed points.

799 **What is proved and what is hypothesised.** Stage 1 is empirical and outcome-defined, stage 2
800 mechanistic, and **neither alone establishes low ASR, nor does their conjunction**: Theorem 1 es-
801 tablishes that C2∧C3 preserve d_2 's discriminative mechanism on the reweighted inputs and **bounds**
802 **no ASR**. That mechanism preservation plus C1 predicts *low ASR* is a separate empirical hypothesis,
803 tested in §5.

804 B The four channels, all four aggregators

805 Endpoint change ($\kappa=0$ against $\kappa=2$) in each measured channel under Mode S, for every
806 downstream defense in the paper. **The first column, Δ agg., is the relative displace-**
807 **ment of the aggregate the defense emits**, $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$ —what enters
808 training, and **not** the value of a scoring statistic. The measured channels are therefore ag-
809 gregate displacement, decision, admission and influence. **Recomputed, not transcribed:**
810 every cell is emitted by experiments/build_channel_table.py from admission_
811 measurement.json, femnist_admission.json, targeted_dose/summary.json, dose_
812 replication/summary.json and dose_femnist/summary.json under results/, which the
813 script only reads; it also asserts that the identity rung displaces nothing (0.00e+00 for all five
814 rows). The influence column is **aggregator-specific by necessity**—a selector has no weights—and
815 its instantiation is: adversarial share of the Weiszfeld/reputation weight mass for the averager,
816 adversarial share of selected coordinates for the order statistic, and the indicator that an adversarial
817 update is selected at all for the two selectors. The FEMNIST row is the second-dataset replication
818 of §5.3; its channels were measured prospectively, before its ASR existed.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (FEMNIST)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017

820 **What this table does and does not show.** It shows that the dissociation between the decision
821 channel and the admission channel is not a property of one aggregator: it holds for a selector, for a
822 cosine-invariant selector, for a coordinate-wise order statistic and for a weighted averager, on two
823 datasets. It does **not** order the arms— $\Delta\Lambda_a$ and Δ ASR disagree in rank, which is the refutation of §4
824 stated in one row-pair—and it is *not* an out-of-sample test: all five rows are Mode-S ladders reported
825 elsewhere in this paper.

826 C Decomposing the emitted displacement: re-selection against rescaling

827 The Δ agg. column above mixes two things the causal claim needs apart. For a selector, $\text{agg}(U) =$
828 u_{s_0} and $\text{agg}(T(U)) = c_{s_1} u_{s_1}$, with s_0 the client selected on the raw stack and s_1 the client selected
829 on the transformed one, so exactly

$$c_{s_1} u_{s_1} - u_{s_0} = \underbrace{(u_{s_1} - u_{s_0})}_{\text{re-selection}} + \underbrace{(c_{s_1} - 1) u_{s_1}}_{\text{rescaling}},$$

830 both components reported relative to $\|u_{s_0}\|$. They are **not orthogonal**—the measured angle runs
831 from $\cos = -0.99$ to -0.43 , mean ≈ -0.65 —so their norms need not sum to the total, and the
832 table prints the angle rather than leaving that discrepancy unexplained. What is asserted instead
833 is the exact vector identity: worst residual $2.66\text{e-}07$ over all 240 (arm, family, rung, seed, round)
834 cells, float32 noise. **This decomposes the published number, not a second similar one:** every
835 recomputed total agrees with the corresponding Δ agg. value in `admission_measurement.json`
836 to $0.00\text{e}+00$, on the same seeds, rounds and configuration. Selectors only—a weighted averager
837 and a coordinate-wise order statistic emit no single selected client, so the split is undefined for
838 them. `experiments/measure_displacement_decomposition.py` trains no ASR, writes a new
839 file, and leaves every frozen artifact byte-identical.

Design	Selector	κ	total	re-selection	rescaling	cos	sel. chg.
Confounded ladder	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.637	0.685	0.333	-0.65	0.600
		1	0.731	0.851	0.685	-0.66	0.733
		2	0.880	1.675	1.674	-0.72	0.867
Instrument (Mode S)	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.663	0.646	0.308	-0.64	0.533
		1	0.847	1.075	0.666	-0.69	0.800
		2	0.892	0.986	0.973	-0.66	0.733
Confounded ladder	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.270	0.000	0.270	—	0.000
		1	0.515	0.000	0.515	—	0.000
		2	0.834	0.000	0.834	—	0.000
Instrument (Mode S)	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.395	0.000	0.395	—	0.000
		1	0.754	0.000	0.754	—	0.000
		2	1.297	0.000	1.297	—	0.000

841 **What it says, and what it does not.** Mode S into Krum at $\kappa=2$: the published 0.892 is 0.986 of re-
842 selection and 0.973 of rescaling, so the dose moves *which* benign client is aggregated and *how large*
843 it arrives, in comparable measure. Of the 11/15 rounds whose selection changes there, **none** lands on
844 an adversary—the quantitative form of the claim that Krum’s decision changes are benign-to-benign,
845 and the reason admitted adversarial mass stays at exactly 0.000 while the decision channel moves
846 by 0.733. `cos_krum` is the informative contrast: its statistic is exactly invariant to positive rescaling,
847 so 0/15 selections change at every rung of *both* designs and its entire displacement—1.297 under
848 the instrument, 0.834 under the ladder—is rescaling. Its fall therefore cannot travel through the
849 decision channel at all, which corroborates §5.3’s attribution of that fall to the magnitude channel
850 from a quantity measured independently of any ASR. **This is a decomposition of a displacement,**
851 **not of an ASR effect:** it says what the dose moves in the update entering training and makes no
852 claim about suppression, and it is an analysis of runs already reported rather than a new experiment.

853 D Related work, continued

854 The attack side of the same literature engineers *persistence* rather than magnitude: model-
855 scaling [Bagdasaryan et al., 2020] and Neurotoxin [Zhang et al., 2022] both aim at triggers that
856 survive rounds of clean aggregation, which is the quantity we measure as γ (§1) and the reason tem-
857 poral defense switching does not help in our regime (Appendix K). Nothing in this paper proposes
858 a new aggregator, a new attack, or a new defense; the object of study is the *composition* of defenses
859 that already exist, and the contribution is an identification argument about how such compositions
860 can and cannot be evaluated.

861 E Notation

symbol	meaning
c_i	upstream coefficient applied to client i 's update, $T(u_i) = c_i u_i$, $c_i > 0$
A	the set of adversarial participants in a round
ρ	statistic-level dispersion, $\max_i c_i / \min_i c_i$: <i>how far apart</i> the coefficients are
Δ_c	adversary-correlated dispersion, $\mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$: <i>who</i> receives them
862 r	adversary-to-benign weight ratio, $w_a / \max_b w_b$; the Mode-A dial, $r = e^\nu$
S	the attack's coordinate extremeness (Theorem 1(1))
m	RFA separation margin, $\Delta'_{\text{sep}}/2(R'_B + \delta')$
Λ_a	aggregate adversarial influence: adversarial share of the Weiszfeld mass
κ	the Mode-S / ladder dispersion dial, $\rho = e^{2\kappa}$
γ	attack <i>persistence</i> , $\text{ASR}(t+1)/\text{ASR}(t)$ after a clean round (≈ 0.66); never a weight

863 F Deployment guidance in full

864 **Deployment guidance.** (0) Check C0: if one candidate defense already suppresses every attack
865 you have, deploy it alone—composition has nothing to add. (1) Check C1: does at least one defense
866 suppress known attacks? (2) Check C2 by invariance class (Proposition 1): a cosine-similarity down-
867 stream defense has its signal preserved behind any positive rescaling, a consensus-distance one does
868 not. (3) For weighting→aggregator compositions Theorem 1 gives verifiable sufficient conditions
869 for C2∧C3. Gate on C2; read C3 as a *quantitative diagnostic*: a 5–10 round pilot returns ρ and the
870 separation margin m , and it is $S > \rho$ (rank aggregators) or $m \gg 1$ (RFA) that matters—not whether
871 d_1 happens to be heterogeneous, the binary reading that produced both of our false negatives. (4)
872 Best candidate on our evidence, for further evaluation rather than deployment: **FG→CM** (accuracy
873 0.68, max ASR 0.107). FG→RFA is marginally lower on the point estimate (0.093 at $n=30$) but
874 statistically indistinguishable, and carries the *highest* adaptive ASR (0.167) and an 18-point accu-
875 racy cost, so it is not the more secure choice on our evidence. (5) In our persistent-backdoor setting
876 fixed-order composition substantially outperforms randomized ordering; we have not tested regimes
877 where randomization might pay.

878 G Proofs

879 The proofs below are short; they are collected here rather than inline because what each result *rules*
880 *in or out* is the content, not the derivation. Both witnesses of Proposition 4 are runs reported in §4
881 and §5.3: Mode S into Krum for part (a), `cos_krum` for part (b).

882 *Proof of Proposition 2.* The load-bearing condition excludes d_1 from attaining the minimum in (i),
883 so certification forces $M(d_2, a) < \theta$; both cells with $M(d_2, a) \geq \theta$ are unoccupied whatever the
884 value of P . A contrast varying d_2 therefore compares pairs differing in P only inside the selected
885 region $M(d_2, a) < \theta$, where P and $M(d_2, a)$ are both functions of the same choice of d_2 ; the
886 missing cell is the one in which they vary independently, and no reweighting of the occupied cells
887 recovers it. The emptiness follows from the gate rather than from $\mathcal{D}$. $\square$

888 *Proof of Lemma 2.* C1 at a^* gives $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a^*) < 0.5$. By C0 the minimum is not
889 attained at d_1 , so it is attained at d_2 . $\square$

890 *Proof of Proposition 3.* Emergence is the negation of the gate, so each member of $\mathcal{E}$ is rejected while
891 belonging to $\mathcal{S}$: a false negative. The bound is the resulting recall. $\square$

892 *Proof of Proposition 4.* Both halves are exhibited. (a) Mode S into Krum under model-scaling, $\kappa=0$
893 against $\kappa=2$: ρ moves from 1 to 54.6 and Krum's decision changes in up to 0.80 of rounds, while the
894 admitted adversarial mass is exactly 0.000 at every rung, the adversarial coefficient share is constant
895 at 0.266667, and suppression is preserved ($|\Delta|=0.026 < 0.15$). (b) `cos_krum` under the pixel
896 backdoor: by Proposition 1(a) its statistic is invariant, and its decision and admission are measured
897 bit-identical (0.000) at every rung, yet ASR falls 0.173 by the first rung (§5.3). $\square$

898 *Proof of Proposition 5.* By construction: Mode S pins every adversarial coefficient at $c=1$ and
 899 spreads only benign clients, giving $\Delta_c \equiv 0$ at every ρ ; Mode A holds the benign weights uni-
 900 form and sweeps only the adversary-to-benign ratio, moving Δ_c at fixed dispersion. Realized values
 901 are tabulated in Appendix N, computed with no ASR involved. $\square$

902 *Proof of Proposition 1.* (a) $c_i u_i / \|c_i u_i\| = u_i / \|u_i\|$ for $c_i > 0$, so the matrix of pairwise cosines is
 903 unchanged; FoolsGold’s weights are a function of that matrix alone (via each client’s normalized
 904 maximum similarity), hence unchanged. (c) With $u_1=(1, 0)$, $u_2=(0, 1)$, $u_3=(3, 3)$ the coordinate-
 905 wise median is $(1, 1)$ and u_3 is farthest; under $c = (1, 1, 0.1)$ the median becomes $(0.3, 0.3)$ and u_3
 906 becomes the *closest* point, reversing the ordering Reputation depends on. $\square$

907 *Proof of Theorem 1.* (1) CoordMedian operates on $\{w_i u_{i,k}\}_{i=1}^K$ per coordinate. If the adversar-
 908 ial values lie strictly beyond all benign values on one side (ii) and $n_a < K/2$, then at most
 909 $n_a < \lceil K/2 \rceil$ of the extreme order statistics are adversarial, so the median is a benign value. For
 910 the magnitude condition: $w_a > 0$ gives $\rho \geq \max_b w_b / w_a$, so $|u_{a,k}| > \rho \max_b |u_{b,k}|$ implies
 911 $|w_a u_{a,k}| > w_a \rho \max_b |u_{b,k}| \geq (\max_b w_b) \max_b |u_{b,k}| \geq \max_b |w_b u_{b,k}|$; with a common sign this
 912 places $w_a u_{a,k}$ strictly beyond every benign value in that direction. Model-scaling sets $u_a = S \tilde{u}_a$
 913 with $S=10$, which amplifies the adversary’s own direction and so satisfies the sign assumption by
 914 construction; with measured $\rho \leq 2.1$ the margin is $S/\rho \geq 4.8$. TrimmedMean additionally needs
 915 the trimming budget to cover the adversarial count.

916 (2) Every benign residual satisfies $\|u'_b - \mu'\| \leq R'_B + \delta'$. For adversarial u'_a : $\|u'_a - \mu'_B\| \geq \Delta'_{\text{sep}} - R'_B$,
 917 so $\|u'_a - \mu'\| \geq \Delta'_{\text{sep}} - R'_B - \delta'$. Requiring this to exceed $R'_B + \delta'$ gives (i). $\square$

918 Theorem 1(2) orders residuals *per client*, which alone does not bound the adversary’s *total* share of
 919 the aggregate—several adversaries could each be individually down-weighted yet collectively hold
 920 substantial mass. The separation margin controls this, quantitatively:

921 **Corollary 1** (Bounded aggregate adversarial influence). *Let $m := \Delta'_{\text{sep}} / [2(R'_B + \delta')]$ be the realized*
 922 *separation margin, so Theorem 1(2) reads $m > 1$. Benign residuals are at most $D_b := R'_B + \delta'$ and*
 923 *adversarial residuals at least $(2m - 1)D_b$, so bounding $\sum_j 1/d_j$ below by the benign terms alone*
 924 *gives*

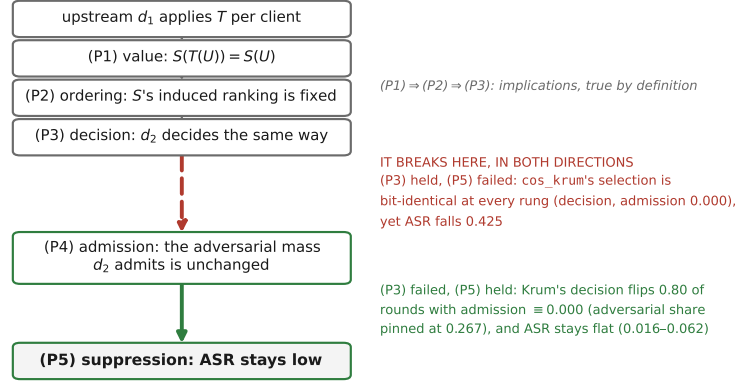
$$\Lambda_a = \sum_a \lambda_a \leq \frac{n_a}{n_b(2m-1) + n_a}.$$

925 *The aggregate is thus a convex combination whose adversarial share is at most Λ_a : writing $\mu' =$*
 926 *$(1 - \Lambda_a)\bar{u}_B + \Lambda_a\bar{u}_A$ for the benign and adversarial parts, the displacement from the benign-only*
 927 *combination $\bar{u}_B$ is at most $\Lambda_a\|\bar{u}_A - \bar{u}_B\|$.*

928 The bound is only as good as the margin: as $m \rightarrow 1$ it degrades to $n_a/(n_b + n_a)$, approaching
 929 40% of the mass at $n_a=2$, $n_b=3$. We measure $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71
 930 over those rounds), capping the adversarial share at 12.6% for $n_a=2$; the realized Λ_a never exceeds
 931 1.8%, and the bound holds in all 17 measured rounds.

932 H Verification of the invariance mirrors

933 All eight metric-swap labels were frozen in advance, and these mirrors are mechanism-
 934 isolating ablations on the same footing as the instruments of §4. We check the statisti-
 935 cally directly rather than inferring invariance from an outcome (verify_cos_invariance.py,
 936 results/cos_invariance_check.json), comparing each d_2 ’s internal state on $\{u_i\}$ against
 937 $\{c_i u_i\}$ for the c_i NormClip and RFA actually produce. Over 18 live rounds (3 seeds $\times$ 2 up-
 938 streams $\times$ 3 rounds) cos_reputation’s weight vector matches to $\leq 2.5 \times 10^{-7}$ with its ordering
 939 never changing (0/18) and cos_krum selects the same client in 18/18; the Euclidean originals do
 940 not (Reputation’s weights move by up to 0.222, ordering changes 12/18; Krum’s selection changes
 941 6/18). Disturbance tracks the realized weight ratio ρ as Proposition 1 predicts: under NormClip
 942 ($\rho \in [1.00, 4.90]$) Reputation changes in 3/9 and Krum in 0/9, under RFA ($\rho \in [2.11, 17.38]$)
 943 in 9/9 and 6/9. On 1000 synthetic draws the separation is cleaner still (0/1000 vs. 992/1000
 944 for the Reputations, 0/1000 vs. 618/1000 for the Krums), and the mirrors match the shipped im-
 945 plementations they stand in for to $\approx 10^{-8}$ relative aggregate error. **The post-hoc within-defense**
 946 **contrast of §5.1 re-uses this frozen suite**, so it is not pre-registered, and two seeds were added
 947 after seeing the $n=3$ result—a top-up that cut both ways, degrading the invariant control’s Δ from



Statistic preservation is not security preservation: only the (P4) $\rightarrow$ (P5) link carries security content in our experiments.

Figure 2: **The preservation chain, and where it breaks.** (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) holds by definition; **the (P3)–(P4) link breaks in both directions**, and both witnesses come from the oracle intervention that pins every adversary’s coefficient and so closes Lemma 1’s attenuation channel *by construction* (§4). The pooled statistic-level prediction we had frozen is separately **rejected** by a pre-registered ladder (§5.2), and a pre-registered replication in a different invariance class reproduces the negative but **refutes the cross-arm admission ordering**, so **admission is a direction, not a validated predictor**. Every number in the figure is recomputed from the scoring scripts’ own JSON by `story_chain_figure.py`.

948 ≈ 0 to $+0.19$ while leaving the disturbed/undisturbed split without overlap. One asymmetry mat-
 949 ters: `cos_reputation`’s *weights* are exactly invariant but its aggregate is not, since the weighted
 950 average is over the rescaled $c_i u_i$ —a magnitude contraction of the Lemma 1 kind, so the prediction
 951 is two-sided, with ASR staying low while clean accuracy pays a contraction cost.

952 I Dose–response caveats recorded before the freeze

953 **The pre-freeze disturbance measurement.** The abscissa was measured over 5 seeds $\times$ 3 rounds
 954 per cell with no ASR involved, which is why it precedes the freeze: the class-(a) mirror’s cosine
 955 scores move by at most 1.1×10^{-4} in 0/120 cells, i.e. the invariance holds in the shipped code
 956 and not merely on paper, while the class-(c) arms reach 0.87 and 0.93. **Attack assignment.** The
 957 class-(c) arms are on model-scaling and the class-(a)/(b) arms on the pixel backdoor: not a free
 958 choice, since on model-scaling *no* invariant d_2 suppresses at usable accuracy (§5.1). “Same at-
 959 tack” is required *within* each ladder, which is where the causal claim lives; across arms we com-
 960 pare curve shapes and slopes, never levels. `coord_median/pixel` sits only 0.057 below threshold,
 961 so its binary above/below-0.5 reading is weak either way. The dose is uncorrelated with adver-
 962 sary status by construction but not adversary-*adversarial*; a dose that deliberately up-weights ad-
 963 versaries would be strictly harder and is not run. A synthetic transform is not a real defense, so
 964 we overlay the two *real* upstreams already measured on these cells (NormClip, RFA) as anchors
 965 at their own measured disturbance; landing far off the synthetic curve would be a finding about
 966 the instrument and reported as one. Design, grid, arms, shapes, margin and refutation criteria
 967 are frozen in `pre_registration_dose_response.md` at commit e711a95, timestamped before
 968 `results/dose_response/` existed and reproduced verbatim in the runner’s arm table. **What it**
 969 **does not address:** recall, and discovery—`fg` $\rightarrow$ `rfa` remains a false negative of our own rule.

970 J Dose–response ladder: per-seed detail, intervals, and the anchor residuals

971 Table 5 is every cell of the ladder; Figure 4 plots it on the measured axis and Figure 5 on the nominal
 972 one. Both are produced by `analyze_dose_response.py` and `dose_response_figure.py` from
 973 `results/dose_response/`; nothing below is transcribed from a log.

Table 2: **The confounded ladder against the Mode-S instrument, on the one arm that has both.** Same downstream defense, same attack, same seeds, same nominal dose—and the two designs share their $\kappa=0$ cell *numerically* (mean and worst per-seed gap 0.00e+00, asserted at 10^{-12}). At the top rung they are matched on aggregate displacement, decision change and mean clean accuracy, so **the ladder’s fall is not an accuracy collapse**; they differ chiefly in the **adversarial coefficient channel**, and the ASR effect **changes sign**. **The one non-match, disclosed:** aggregate adversarial influence Λ_a moves in *both* designs, 0.081 against 0.033 ($2.4\times$), so neither column is a clean statistic-only intervention—which is why the contrast is “differ chiefly in the coefficient channel”, never “only in it”. *Vocabulary:* this row is the $\Delta\Lambda_a$ column of the channel table—the adversarial share of selected coordinates, the same field in the same artifact—and **not** its Δ adm. column, which counts rounds where the *support* of the adversarial mass changes and reads 0.000 on every row: what moves here is how much weight the adversary’s direction carries, not whether it enters at all. A two-column contrast on one arm: it locates where the two designs differ and attaches no p -value to the difference between the columns. Every cell recomputed from per-seed and per-round rows by `build_comparability_table.py`.

	Confounded ladder dose_kappa	Instrument (Mode S) doseS_kappa
downstream d_2	coord_median	coord_median
attack	committed pixel	committed pixel
seeds	42–46	42–46
nominal dose	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6
benign coefficients	mean 1 over <i>all</i> clients	mean 1 over <i>benign</i> clients
adversarial coefficients	drawn from the same spread	pinned $c = 1.0$ exactly
adv. coefficient share	0.267 $\rightarrow$ 0.187	0.266667, constant
Δ_c	+0.000 $\rightarrow$ -0.336	0 by construction (read back 9e-08)
aggregate displacement	0.000 $\rightarrow$ 0.722	0.000 $\rightarrow$ 0.670
decision change	0.000 $\rightarrow$ 0.487	0.000 $\rightarrow$ 0.482
adversarial influence $\Delta\Lambda_a$	0.000 $\rightarrow$ 0.081	0.000 $\rightarrow$ 0.033
mean clean accuracy	0.767 $\rightarrow$ 0.690	0.767 $\rightarrow$ 0.698
ASR	0.443 $\rightarrow$ 0.170	0.443 $\rightarrow$ 0.541
ASR effect	-0.272 ($p_\downarrow = 0.0015$)	+0.098 ($p_\uparrow = 0.069$)

974 **The replication arm’s post-hoc seed-matched difference.** The frozen primary for the
975 coord_median replication (§4) is the *unpaired* $\Delta = +0.098$, and it is not superseded by what
976 follows. Between-seed variance on this cell is large (identity ASR spread 0.287–0.677), which the
977 unpaired test cannot see through, so we also report the seed-matched difference: positive in 5/5
978 seeds, paired $t = 3.39$, one-sided $p = 0.014$. **This is post-hoc and was not pre-registered;** we
979 report it because it is the honest description of a small consistent effect, and we label it post-hoc
980 wherever it appears. It does not change the frozen verdict—the admission *ordering* is refuted either
981 way.

982 **The primary test, and two post-hoc sensitivities that cannot change its verdict.** The frozen
983 primary test is a one-sided Spearman correlation of *rise* against *measured disturbance* pooled over
984 all 16 cells, confirmed only at $\rho_s > 0$ with $p < 0.05$: we obtain $\rho_s = +0.351$, $p = 0.091$, so it
985 is **refuted**. Two sensitivities are recorded because they are informative about *why*, and are labelled
986 post-hoc because neither was frozen. (i) The four $\kappa=0$ cells are constructed (0,0) points—zero
987 disturbance and zero rise by definition—and dropping them gives $\rho_s = +0.780$, $p = 0.001$ ($n=12$);
988 so the pooled correlation is carried by the arm-2 rise and destroyed by the two arms whose ASR
989 falls, not by the identity anchors. (ii) Against the *nominal* dose κ (i.e. $\log \rho$) rather than measured
990 disturbance, $\rho_s = -0.354$, $p = 0.911$ —the wrong sign, which is the same fact as three arms falling.
991 Neither is scored.

992 **Per-arm trend tests.** One-sided Jonckheere–Terpstra across the four ordered rungs, 5 seeds each,
993 $\alpha = 0.05$; the normal-approximation p agrees with a 20,000-draw permutation p to three decimals
994 throughout. Arm 1: $J = 70$, $z = -0.34$, $p_\uparrow = 0.632$. Arm 2: $J = 144$, $z = +4.64$, $p_\uparrow =$
995 1.7×10^{-6} . Arm 3: $J = 29$, $z = -3.10$, $p_\uparrow = 0.999$ ($p_\downarrow = 9.8\times 10^{-4}$); dropping the gated $\kappa=2.0$

Table 3: Mode S (statistic-only), $n=5$ per arm ($n=8$ for `cos_krum`): mean ASR with the two measured channels. **The decision channel moves far more than ASR throughout.** [†] below the 0.35 accuracy gate. [‡] the replication arm, rank-discordant by design, frozen separately: $\Delta=+0.098$ is below reputation’s $+0.178$, so the cross-arm admission *ordering* is refuted while the negative replicates. Verdicts otherwise scored against readings frozen in advance; per-seed values in Appendix J.

arm	channel	$\rho=1$	2.72	7.39	54.6	test	verdict
krum / scaling	ASR	0.062	0.016	0.023	0.036	$p_{\uparrow}=0.70$	H-admission confirmed, H-stat. refuted
	decision	0.000	0.533	0.800	0.733		
	admission	0.000	0.000	0.000	0.000		
reputation / scaling	ASR	0.017	0.019	0.025	0.195	rise $+0.178$ vs. ladder $+0.762$ ($p=0.0017$)	H-admission confirmed (mostly atten.)
	decision	0.000	0.933	1.000	1.000		
	admission	0.000	0.002	0.008	0.020		
cos_krum / pixel	ASR	0.584	0.411	0.295	0.159 [†]	$p_{\downarrow}=8\times 10^{-4}$ $ \Delta =0.425$	refuted (Prop. 1 practical)
	decision	0.000	0.000	0.000	0.000		
	admission	0.000	0.000	0.000	0.000		
coord_median / pixel [‡]	ASR	0.443	0.442	0.490	0.541	rise $+0.098$ < $+0.150$ ($p_{\uparrow}=0.069$)	ordering refuted; negative repl.
	decision	0.000	0.232	0.358	0.482		
	admission	0.000	0.005	0.013	0.033		

Table 4: The ladder as measured, $n=5$, per-attack mean ASR (mean clean accuracy). *Predicted* shapes were frozen before any run (Appendix S); $p_{\uparrow}$ is the one-sided Jonckheere–Terpstra probability of an increasing trend, the only direction any frozen rule tests. An arm meeting neither its confirmation nor its refutation criterion is reported *indeterminate* rather than scored in our favour. [†] below the 0.35 accuracy gate: a collapsed model, not suppression. Intervals, per-seed values, anchor residuals and both plotted axes: Appendix J.

arm (d_2 / attack)	predicted shape	mean ASR (clean acc.)				JT $p_{\uparrow}$	verdict
		$\kappa=0$	0.5	1.0	2.0		
1 krum / scaling	monotone rise	0.062 (0.57)	0.011 (0.65)	0.014 (0.68)	0.054 (0.62)	0.63	refuted
2 reputation / scaling	monotone rise	0.017 (0.78)	0.022 (0.78)	0.487 (0.77)	0.779 (0.70)	1.7×10^{-6}	confirmed
3 cos_krum / pixel	FLAT (control)	0.490 (0.61)	0.450 (0.57)	0.153 (0.47)	0.086 (0.22) [†]	0.999	indeterminate
4 coord_median / pixel	rise, shallower	0.443 (0.77)	0.428 (0.76)	0.327 (0.75)	0.170 (0.69)	0.998	refuted

996 rung leaves $z = -2.06$, $p_{\downarrow} = 0.020$, so the decrease is not an artifact of the collapsed cell alone.
997 Arm 4: $J = 31$, $z = -2.96$, $p_{\uparrow} = 0.998$ ($p_{\downarrow} = 1.5\times 10^{-3}$). No frozen rule tests the decreasing
998 direction; $p_{\downarrow}$ is descriptive.

999 **Clean accuracy against dose separates two ways of losing ASR.** Descriptive, not pre-registered.
1000 Accuracy trends by the same test: arm 1 $z = +1.35$, arm 2 $z = -3.30$, arm 3 $z = -3.50$, arm
1001 4 $z = -4.11$. Arm 3 pays $0.607 \rightarrow 0.220$ —a 0.39 collapse through the Lemma 1 contraction
1002 channel, and the reason its $\kappa=2.0$ cell is uninterpretable rather than a success. Arm 4 pays only
1003 $0.767 \rightarrow 0.690$ while ASR more than halves, so there the dose genuinely *helps* coordinate median:
1004 a refutation of our predicted shape, but a benign mechanism. Arm 2 is the cleanest cell in the ladder,
1005 losing 0.076 accuracy while ASR rises $46\times$.

1006 **Anchor residuals, and why the instrument is the finding.** Residual is real-transform ASR
1007 ($n=5$, from `metric_swap_topup`) minus the synthetic rung at the nearest measured distur-
1008 bance. NormClip→Krum: disturbance 0/9, ASR 0.062 vs. rung 0.062, residual $+0.001$.
1009 RFA→Krum: 6/9 = 0.667, ASR 0.779 vs. rung 0.011 at disturbance 0.600, residual
1010 $+0.767$. NormClip→Reputation: 3/9 = 0.333, ASR 0.806 vs. rung 0.017, residual $+0.789$.
1011 RFA→Reputation: 9/9, 0.895 vs. 0.779, residual $+0.116$. Both `cos_krum` anchors sit at distur-
1012 bance 0 and land ($+0.000$, -0.044). `coord_median` was not in the metric-swap menu, so arm 4
1013 has *no* real-transform anchor and its ecological validity is untested. The two large residuals are not
1014 a range mismatch: realized ρ is mean 3.61 / max 4.90 for NormClip and mean 11.93 / max 17.38
1015 for RFA, both inside the ladder’s [1.00, 54.60] span. What differs is *which* clients get reweighted—
1016 NormClip and RFA reweight in a way correlated with adversary status, our permutation deliberately
1017 does not—and we state that as the hypothesis caveat (iii) was written to test, not as a measured quan-
1018 tity: the shipped disturbance logs record only $c_{\min}$, $c_{\max}$ and ρ per round, not per-client coefficients,
1019 so an adversary-versus-benign coefficient contrast requires a new (cheap, no-ASR) measurement.

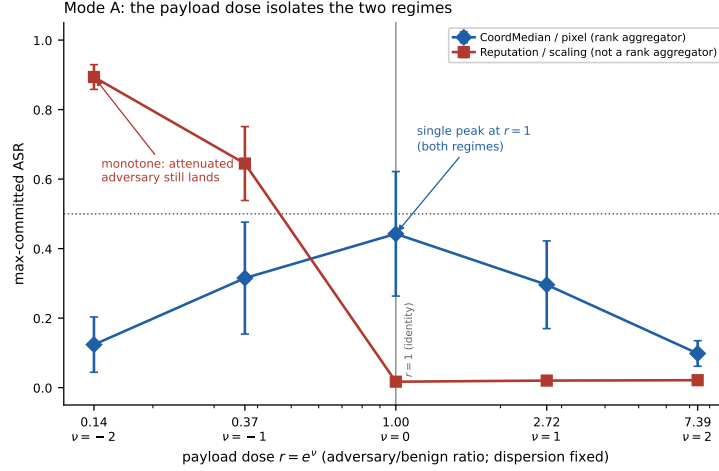


Figure 3: **The two regimes, exhibited: the payload dose separates them.** *The picture of regimes A and B of §2, produced by the instrument of §5.3.* With the statistic’s dispersion fixed and only the adversary/benign weight ratio r swept, `coord_median` (a rank aggregator, the class the regimes are stated for) is single-peaked at $r=1$ as Theorem 1 predicts, while reputation is monotone—off-spec, and the theorem is not claimed for it. Bars are Student- t 95% intervals, recomputed per seed by `analyze_targeted_dose.py`; the Mode-S companion is Table 3.

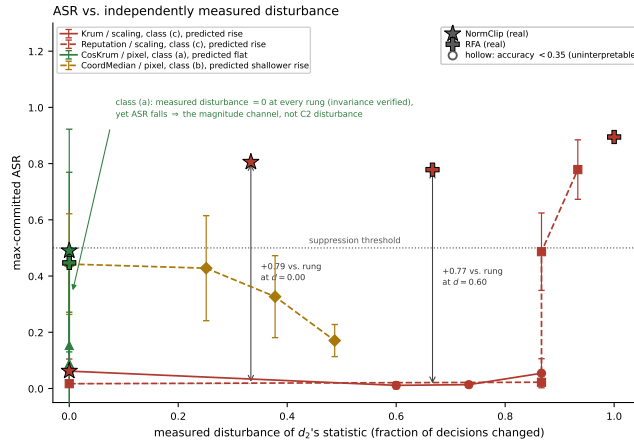


Figure 4: **The measured axis: ASR against disturbance of each arm’s own statistic**, an abscissa fixed before the freeze ($n=5$, Student- t 95% intervals; the figure script imports the arm and anchor construction from the scoring script, so the two cannot diverge). Stars and crosses are the two *real* upstream transforms measured on these same cells: at matched disturbance they sit +0.77 and +0.79 ASR *above* the synthetic curve, though their realized weight ratios (mean 3.61 and 11.93, maximum 4.90 and 17.38) lie *inside* the ladder’s span—§5.2 attributes the gap to Δ_c , post-hoc. The class-(a) arm’s four rungs coincide at disturbance exactly 0 and its ASR falls anyway—Lemma 1’s magnitude channel, not C2 disturbance. Hollow markers fail the 0.35 accuracy gate.

1020 **Two checks that passed.** The $\kappa=0$ rung reproduces each arm’s published standalone value exactly
1021 on every shared seed (16 seed-level comparisons across four arms, largest $|\Delta \text{ASR}|$ and $|\Delta \text{acc}|$ both
1022 $0.00\text{e}+00$ at tolerance 10^{-6}), so $\kappa=0$ really is the identity and the fresh rung and the published
1023 baseline are the same quantity. And `cos_krum`’s selection never changed in any of the 120 measured
1024 cells, with maximum relative score drift 1.1×10^{-4} —Proposition 1’s classification of which statistics
1025 a positive per-client rescaling can disturb is verified directly, independently of the outcome the ladder
1026 then failed to predict.

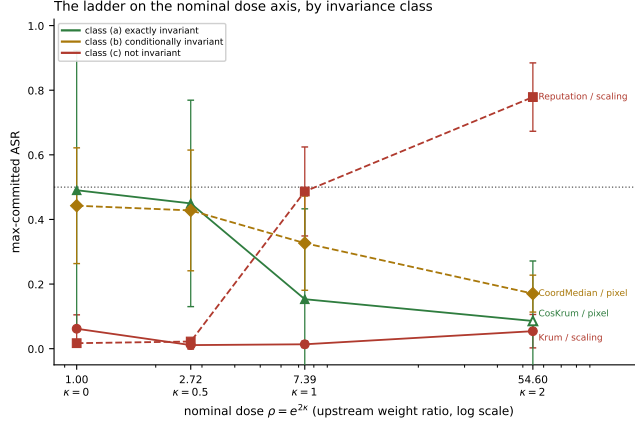


Figure 5: **The nominal axis:** the same 16 cells against the dose $\rho = e^{2\kappa}$, coloured by Proposition 1 class. Of the two class-(c) arms the fragility claim is stated over, only reputation rises; the other two arms *fall*. This is the axis the frozen per-arm shapes were stated on, and the pooled prediction was stated on the measured axis of Figure 4. Hollow markers fail the 0.35 accuracy gate.

Table 5: All 16 rungs, $n=5$. *disturb* is the independently measured disturbance of that arm’s own statistic at that rung, computed before the freeze with no ASR involved; *rise* is ASR relative to the arm’s own $\kappa=0$ rung and is the primary test’s ordinate. CIs are Student- t at $n=5$ and are wide by design: the ladder was powered to detect a monotone trend, not to resolve single cells. † fails the 0.35 clean-accuracy gate.

arm	κ	ρ	dist.	ASR	95% CI	rise	acc	per-seed ASR (42–46)							
krum scaling class (c)	0	1.00	0.000	0.062	[0.019, 0.105]	+0.000	0.565	0.115	0.021	0.047	0.057	0.069			
	0.5	2.72	0.600	0.011	[−0.001, 0.023]	−0.051	0.652	0.000	0.010	0.011	0.027	0.007			
	1.0	7.39	0.733	0.014	[0.003, 0.024]	−0.048	0.682	0.003	0.015	0.021	0.021	0.008			
	2.0	54.60	0.867	0.054	[0.002, 0.106]	−0.008	0.619	0.005	0.039	0.077	0.113	0.037			
reputation scaling class (c)	0	1.00	0.000	0.017	[0.012, 0.022]	+0.000	0.777	0.014	0.012	0.017	0.023	0.019			
	0.5	2.72	0.867	0.022	[0.014, 0.031]	+0.005	0.779	0.016	0.018	0.024	0.033	0.020			
	1.0	7.39	0.867	0.487	[0.349, 0.624]	+0.470	0.765	0.439	0.547	0.578	0.557	0.313			
	2.0	54.60	0.933	0.779	[0.673, 0.885]	+0.762	0.701	0.710	0.843	0.730	0.896	0.714			
cos_krum pixel class (a)	0	1.00	0.000	0.490	[0.058, 0.922]	+0.000	0.607	0.003	0.269	0.630	0.864	0.686			
	0.5	2.72	0.000	0.450	[0.130, 0.769]	−0.041	0.569	0.764	0.236	0.165	0.648	0.435			
	1.0	7.39	0.000	0.153	[−0.127, 0.433]	−0.337	0.466	0.132	0.090	0.000	0.543	0.001			
	2.0	54.60	0.000	0.086	[−0.100, 0.271]	−0.405	0.220 †	0.083	0.000	0.000	0.345	0.000			
coord_median pixel class (b)	0	1.00	0.000	0.443	[0.263, 0.622]	+0.000	0.767	0.395	0.287	0.449	0.677	0.405			
	0.5	2.72	0.251	0.428	[0.241, 0.615]	−0.015	0.764	0.454	0.215	0.430	0.638	0.402			
	1.0	7.39	0.378	0.327	[0.181, 0.472]	−0.116	0.749	0.351	0.139	0.352	0.463	0.328			
	2.0	54.60	0.487	0.170	[0.113, 0.228]	−0.272	0.690	0.179	0.090	0.204	0.192	0.187			

1027 K Secondary results: baselines, the adaptive stress test, scale, and switching

1028 **The margin behaves like a certificate, not just a switch.** C3 is binary, but the margin m is
1029 measurable and its *strength* tracks security: over the 29 measured rounds with $w_a > 0$, higher
1030 margin goes with lower realized adversarial mass (Spearman $\rho = -0.48$, $p = 0.009$), Corollary 1’s
1031 bound holding in every one—a number rather than a verdict, though the link is directional, not fitted,
1032 and post-hoc.

1033 **Single-defense baselines.** Measured under the identical protocol (5 seeds, 50 rounds), no individ-
1034 ual defense achieves low max-committed ASR: CM 0.52, FG 0.73, Rep 0.84, TM 0.85, RFA 0.89,
1035 NC 0.94. Against the best single defense (CM, 0.52), FG→RFA (0.093) is $5.6\times$ lower and FG→CM
1036 (0.107) is $4.9\times$ lower. FoolsGold alone reaches ASR 0.20 on model-scaling only by collapsing the
1037 model to 10% accuracy, so it is not suppression—consistent with its C1 failure.

1038 **Adaptive attacks against FG→RFA.** We evaluate one whitebox adaptive attack family that
1039 jointly evades both defenses: disjoint coordinate blocks minimize cosine similarity (evade FG) while

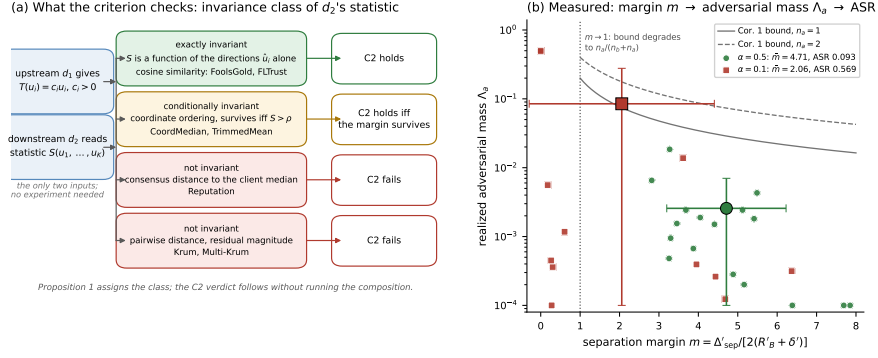


Figure 6: **The criterion is a mechanism test, and the mechanism is measurable.** (a) What C2 checks: the upstream transform T and the statistic S that d_2 reads determine an *invariance class* (Proposition 1), and the class determines the verdict with no composition run—which is why norm clipping does not blind FoolsGold but does blind Reputation. (b) The measured chain for FG→RFA, per round, on the transformed points RFA actually receives; each marker is one round with $w_a > 0$ (rounds where FoolsGold zeroes every adversary are Lemma 1 and carry no payload), large markers are mean $\pm$ s.d., and values below 10^{-4} are drawn at 10^{-4} . Corollary 1’s bound holds in every round, and the $\alpha=0.1$ collapse is visible in the theorem’s own quantity: m falls from 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$), with several rounds at $m < 1$ where the bound degrades to $n_a/(n_b+n_a)$, and realized Λ_a rises $0.003 \rightarrow 0.085$ (max 0.50) as composition ASR goes $0.093 \rightarrow 0.569$.

Table 6: Framework validation (representative pairs; full tallies in text). Threshold: LOW < 0.5 , HIGH ≥ 0.5 .

Composition	Category	Predicted	Max ASR	Correct?
<i>PASS: C1∧C2∧C3 (predicted LOW)</i>				
FoolsGold $\rightarrow$ RFA	PASS	LOW	0.093	✓
FoolsGold $\rightarrow$ CoordMedian	PASS	LOW	0.107	✓
Reputation $\rightarrow$ CoordMedian	PASS	LOW	0.423	✓
<i>C1-FAIL: no constituent suppresses (predicted HIGH)</i>				
Reputation $\rightarrow$ RFA	C1-FAIL	HIGH	0.825	✓
NormClip $\rightarrow$ FoolsGold [†]	C1-FAIL	HIGH	0.887	✓
RFA $\rightarrow$ FoolsGold [†]	C1-FAIL	HIGH	0.914	✓
<i>C2-FAIL: discriminative property destroyed (predicted HIGH)</i>				
NormClip $\rightarrow$ Reputation	C2-FAIL	HIGH	0.819	✓
<i>C3-FAIL: margin compressed (predicted HIGH)</i>				
FoolsGold $\rightarrow$ TrimmedMean	C3-FAIL	HIGH	0.508	✓
NormClip $\rightarrow$ RFA	C3-FAIL	HIGH	0.853	✓
<i>False negatives (predicted HIGH, measured LOW)</i>				
NormClip $\rightarrow$ CM / RFA $\rightarrow$ CM	C3-FAIL	HIGH	0.433 / 0.485	×

[†]Reclassified from C2-FAIL: by Proposition 1(a) these preserve C2.

1040 ϵ -ball projection stays near the geometric median (evade RFA). The adversary has full state access,
1041 and these constraints are in tension. ASR against FG→RFA (5 seeds each, base quoted on the *same*
1042 five seeds so the comparison is seed-matched rather than against the $n=30$ mean): base committed
1043 0.051 ± 0.011 ; Neurotoxin [Zhang et al., 2022] top-20% at scale 2.0 0.048 ± 0.009 ; projection only,
1044 no decorrelation, $\epsilon=2$: 0.050 ± 0.008 . With joint constraints the sweep over ϵ gives 0.036 ± 0.015
1045 ($\epsilon=0.5$), **0.167 ± 0.069** ($\epsilon=1$), 0.130 ± 0.030 ($\epsilon=2$), 0.124 ± 0.025 ($\epsilon=4$). ASR peaks at $\epsilon=1$:
1046 tighter projection over-constrains the attack, larger ϵ fails to evade RFA, and without decorrelation
1047 FoolsGold suppresses it outright, so the competing constraints leave a narrow effective window.
1048 This is an empirical finding—Theorem 1 concerns mechanism preservation and does not bound an
1049 adaptive adversary whose optimization is untied to its conditions—and it probes one natural family

of joint constraints at a handful of ϵ values, not the adaptive attack space. Within it, 0.167 is $3.3\times$ the seed-matched committed-attack ASR ($1.8\times$ the $n=30$ mean).

Scale. At $N=100$, $f=0.05$, ResNet18 both pre-registered PASS predictions hold: $\text{FG}\rightarrow\text{RFA}$ max ASR = 0.069, $\text{FG}\rightarrow\text{CM}$ = 0.152. We flag that these runs reach only ≈ 0.29 clean accuracy at 50 rounds—ResNet18 is far from converged at this budget—so they establish that suppression transfers across scale and architecture, not that the composition is deployable at $N=100$ as configured.

K.1 Heterogeneity, and where the criterion breaks

Earlier drafts swept heterogeneity for single defenses only; no composition had been run at any $\alpha \neq 0.5$. We therefore ran all three PASS pairs across $\alpha \in \{0.1, 1.0, 10.0\}$ (3 seeds, both attacks, 50 rounds; max-committed ASR / clean accuracy):

Composition	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	$\alpha=10.0$
FG $\rightarrow$ RFA	0.569 / 0.20	0.093 / 0.50	0.051 / 0.57	0.063 / 0.62
FG $\rightarrow$ CM	0.433 / 0.40	0.107 / 0.68	0.133 / 0.74	0.129 / 0.78
Rep $\rightarrow$ CM	0.548 / 0.48	0.423 / 0.76	0.377 / 0.79	0.326 / 0.81

In our sweep, which is three seeds per cell and fixes direction rather than magnitude, the certified compositions stay below threshold for $\alpha \in \{0.5, 1.0, 10.0\}$, including the near-IID setting where one might expect FoolsGold to founder on legitimately similar benign updates: $\alpha=10$ is indistinguishable from $\alpha=0.5$. **Under extreme heterogeneity ($\alpha=0.1$) two of the three PASS compositions cross the threshold** (FG $\rightarrow$ RFA 0.569, Rep $\rightarrow$ CM 0.548) and only FG $\rightarrow$ CM survives, at 0.433—a real boundary condition on the framework, with caveats in both directions: FG $\rightarrow$ RFA at $\alpha=0.1$ is strongly bimodal across seeds (0.065–0.923), so at $n=3$ the direction is clear but the magnitude is not, and clean accuracy degrades badly (0.20–0.48), so the regime stresses utility as well as security. Crucially Theorem 1 supplies a diagnostic quantity that retrospectively explains this boundary: re-measuring the transformed points at $\alpha=0.1$, RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%)—over the payload-carrying rounds alone, the ones Figure 6 plots, the same collapse reads 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$) with the per-round minimum falling $2.82 \rightarrow 0.00$, the difference being that the all-round means average in the Lemma 1 rounds at $m=0$ —exactly the boundary $m \rightarrow 1$ where Corollary 1’s bound degrades to 40% adversarial mass, because extreme heterogeneity inflates the benign radius R'_B until the benign cluster is no longer separated from the adversarial points, which is the precondition the theorem requires. A practitioner running the 5–10 round pilot we recommend would measure $m \approx 1$ and decline to deploy. The criterion is therefore mechanistically diagnostic at this boundary—the property that makes a screening rule usable—though this is a post-hoc explanation of a known failure, not a pre-registered prediction of it.

K.2 Composition vs. randomized switching

High persistence sharply reduces the value of temporal switching. In our tested regime, $\gamma \approx 0.66$ ($\gamma := \text{ASR}(t+1)/\text{ASR}(t)$ after one clean round): triggers decay slowly enough that intermittent suppression cannot clear them, so temporal mixing approaches the weaker constituent defense’s performance. Whether switching retains any value depends jointly on γ , injection strength, switching frequency, and recovery dynamics—persistence alone does not imply switching is worthless. The quantitative boundary is architecture-specific; the qualitative prediction—high- γ attacks blunt switching—is what this secondary analysis contributes, and nothing in the preservation argument rests on it.

Order-sensitivity control. Both variants apply both defenses every round and differ *only* in order, so compute is matched and any gap is structural. We report it as a **motivating demonstration that composition order is structurally consequential, not as evidence for the criterion**: the mechanism behind the largest gap is a degeneracy rather than a signal-preservation failure, as the diagnosis below makes clear. Pixel ASR, 5 seeds: fixed FG $\rightarrow$ CM 0.076 ± 0.035 versus randomized 50/50 ordering 0.681 ± 0.133 ($9\times$) and temporal mixing of one defense per round 0.679 ± 0.159 ; fixed FG $\rightarrow$ RFA 0.037 ± 0.011 versus randomized 0.719 ± 0.113 ($19\times$). The mechanism is the

1097 degeneracy of §1: CoordMedian supplies no per-client transform, so every CM→FG round reduces
 1098 to FoolsGold alone, leaving half the rounds defended by a single constituent that cannot suppress
 1099 on its own. On a reputation+TrimmedMean menu ($n=30$), composition reaches ASR 0.316 against
 1100 0.915 for temporal mixing.

1101 L The metric-swap suite’s two-sided prediction

1102 Because C1 binds for all eight pairs, every frozen label in the metric-swap suite was HIGH—a
 1103 consequence of the measured inputs rather than a later choice—so the suite bears on specificity
 1104 only. One prediction inside it was genuinely two-sided. `cos_krum`’s C1 verdict on model-scaling
 1105 turns on whether the accuracy gate is applied (0.078 ASR at 0.146 clean accuracy), so we froze *both*
 1106 readings in advance: primary (gated) $\Rightarrow$ HIGH, secondary (numeric only) $\Rightarrow$ LOW, with the rule
 1107 fixed beforehand that a LOW outcome at accuracy ≥ 0.35 would mean the gate was wrong. Both
 1108 cells came out **HIGH** (0.508, 0.566), so the gate is vindicated against a pre-registered alternative
 1109 that would have overturned it. This is the only place in the paper where our own accuracy gate was
 1110 put at risk by a frozen rule.

1111 M The prospective suite, arm by arm

1112 (i) **FLTrust as d_1 (7 pairs, 2/7): out of scope.** When d_1 alone suppresses, the composition in-
 1113 herits that suppression whatever d_2 does, so all five predicted-HIGH pairs are misses even though
 1114 their C2/C3 failures are real—FLTrust genuinely destroys Reputation’s consensus-distance signal,
 1115 but destroying a mechanism that was never load-bearing costs nothing. This is C0, and it was invis-
 1116 ible on the original 42 pairs, where the strongest single defense reaches only 0.519 and the lowest
 1117 ASR among all 39 non-PASS pairs is 0.433; no pair was ever out of scope. We apply C0 here
 1118 *retrospectively*, which is weaker than a pre-registered exclusion.

1119 (ii) **FLTrust as d_2 (6 pairs, 6/6): in scope, but nearly an identity.** Proposition 1(a) predicts
 1120 $\cdot \rightarrow$ FLTrust preserves C2, and it does—but FLTrust gates on a scale-invariant cosine and then *renor-*
 1121 *malizes every update to $\|g_{\text{srv}}\|$* , discarding magnitude outright, so a positive rescaling $u_i \mapsto c_i u_i$
 1122 is *exactly* cancelled: FLTrust discards precisely what d_1 modifies. The per-seed data show this di-
 1123 rectly. $\text{Krum} \rightarrow \text{FLTrust}$ and $\text{MKrum} \rightarrow \text{FLTrust}$ are **bit-identical** to FLTrust alone on all six runs
 1124 (pass-through d_1); $\text{NC} \rightarrow \text{FLTrust}$ is bit-identical *on the pixel attack*, where no client exceeds $\tau=5$
 1125 so $c_i \equiv 1$, and differs only within seed noise on model-scaling, where clipping fires; $\text{Rep} \rightarrow$ and
 1126 $\text{RFA} \rightarrow \text{FLTrust}$ likewise differ only within noise. The lone real deviation is $\text{FG} \rightarrow \text{FLTrust}$ (0.064,
 1127 up to 0.084 per seed)—the one d_1 producing exact *zeros* rather than positive rescalings, changing
 1128 FLTrust’s client set instead of its magnitudes, which is the Lemma 1 interaction. So this arm con-
 1129 firms that theory and implementation agree, and demonstrates the degeneracy mechanism cleanly,
 1130 but it is **not falsifiable**: every d_1 in our menu is a *non-negative* per-client rescaling, FLTrust cancels
 1131 the positive part exactly, and so the only channel left open is the *zero pattern*—which is precisely
 1132 where the one deviation appears. Proposition 1(a) permits no other outcome on the strictly-positive
 1133 part, and Lemma 1 accounts for the rest.

1134 (iii) **Neither defense is FLTrust (7 pairs, 5/7): the only arm that could have surprised us.**
 1135 All were predicted HIGH; correct are $\text{rep} \rightarrow \text{krum}$ 0.880, $\text{nc} \rightarrow \text{mkrum}$ 0.943, $\text{rfa} \rightarrow \text{mkrum}$ 0.875,
 1136 $\text{krum} \rightarrow \text{rfa}$ and $\text{mkrum} \rightarrow \text{rfa}$ both 0.870. Neither miss is a mechanism error: $\text{fg} \rightarrow \text{krum}$ reads 0.078
 1137 at 9.9% clean accuracy—a collapsed model, the same hollow-suppression artifact as FoolsGold
 1138 alone—and $\text{krum} \rightarrow \text{cm}$ reads 0.450 against a 0.5 threshold as a degenerate pair reducing to Coord-
 1139 Median alone, whose own baseline is 0.519: threshold noise on a borderline constituent. This arm
 1140 contains *zero* predicted-LOW pairs, so like Tier 2 it bears on specificity only.

1141 **The suggestive contrasts, and why they are still confounded.** Three matched pairs hold d_1 fixed
 1142 while varying only d_2 ’s invariance class: $\text{rep} \rightarrow \text{fltrust}$ 0.046 vs. $\text{rep} \rightarrow \text{krum}$ 0.880; $\text{nc} \rightarrow \text{fltrust}$ 0.046
 1143 vs. $\text{nc} \rightarrow \text{mkrum}$ 0.943; $\text{rfa} \rightarrow \text{fltrust}$ 0.047 vs. $\text{rfa} \rightarrow \text{mkrum}$ 0.875—a 19–20 $\times$ gap turning on exactly
 1144 the property Proposition 1 identifies. They are nonetheless confounded in the way Lemma 2 pre-
 1145 dict: the invariant-signal d_2 is also the only d_2 in this menu that suppresses alone, so effectiveness
 1146 and mechanism co-vary across the contrast. This is why breaking the confound requires holding
 1147 standalone effectiveness fixed rather than choosing a different defense.

1148 N Adversary-correlated dispersion Δ_c , measured

1149 $\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ for every per-client transform in the paper,
 1150 over the same 5 seeds $\times$ 3 rounds and, within a round, the *same* raw update stack for every
 1151 transform, so the rows are comparable by construction (measure_coefficient_targeting.py,
 1152 results/coefficient_targeting.json). **No ASR is computed and no model is trained per**
 1153 **transform**, which is what lets this be added without rescoring anything; the figures are **post-hoc and**
 1154 **exploratory** and no frozen prediction depends on them. Coefficients are read back from transformed
 1155 norms, $c_i = \|T(u_i)\|/\|u_i\|$, so Table 7 measures the shipped code path rather than the builder’s
 1156 description of it. As a provenance check the synthetic families’ adversarial coefficient shares repro-
 1157 duce results/admission_measurement.json to 0.0 absolute on all 13 cells, so these rows may
 1158 be read on the same axis as the Mode-S and Mode-A figures of §4.

Table 7: ρ says how much a transform spreads the coefficients; Δ_c says who receives the spread. They are not ordered together: RFA and NormClip target the adversary 2–3.6 $\times$ harder than the ladder’s most dispersed rung at ρ that is 3.7–13 $\times$ *smaller*. Means over non-degenerate rounds; the synthetic rows are attack-pooled since their coefficients do not depend on the attack.

transform	attack	ρ	$\mathbb{E}[c \mid A]$	$\mathbb{E}[c \mid \neg A]$	Δ_c
NormClip	model-scaling	4.19	0.268	0.989	−0.721
NormClip	pixel	1.00	1.000	1.000	0.000
Reputation	model-scaling	8.3×10^5	0.003	1.387	−1.383
Reputation	pixel	1.68	0.915	1.019	−0.105
FoolsGold	model-scaling	1.9×10^{12}	1.497	0.834	+0.663
FoolsGold	pixel	1.8×10^{12}	1.365	0.986	+0.378
RFA	model-scaling	14.8	0.125	1.332	−1.208
RFA	pixel	1.69	0.923	1.012	−0.089
<i>Synthetic families of §5.2–§4, same code path:</i>					
ladder, $\kappa=0.5/1.0/2.0$	—	2.72/7.39/54.6	—	—	−0.140/−0.240/−0.336
Mode S, all rungs	—	1.00–54.6	—	—	0.000
Mode A, $\nu=-2 \dots +2$	—	7.39–7.39	—	—	−1.138 $\dots$ + 2.469

1159 Three readings, in decreasing confidence. **(1) ρ is not a proxy for Δ_c :** the two are anti-ordered
 1160 on the anchors, and NormClip on the pixel backdoor has $\rho = 1$ exactly because no client exceeds
 1161 $\tau=5$, so $\Delta_c = 0$ there for a reason the invariance classification cannot see. **(2) The two anchors**
 1162 **that miss the ladder are the two most extremely targeted**, which explains the residuals of §5.2
 1163 without appeal to a range mismatch. **(3) Δ_c is nonetheless not a predictor of suppression**—RFA
 1164 is the most negative of the four real transforms and rfa→krum fails at 0.779, while Reputation is
 1165 more negative still and rep→coord_median suppresses. That is the two-regime structure of §2 again:
 1166 strong negative Δ_c can mean either annihilation or mere dilution, and Δ_c ’s magnitude alone does
 1167 not say which. We report it as the axis on which the ladder’s anchors were unmatched, not as the
 1168 quantity that replaces the statistic.

1169 O Confusion matrix in full

1170 Exactly 5 of the 42 pairs have max-committed ASR < 0.5 : the 3 PASS pairs plus the 2 false
 1171 negatives. Pooling both in-distribution tiers—each pair scored against the label committed for it
 1172 at the time—gives TP = 3, FP = 2, TN = 35, FN = 2 (38/42; specificity 94.6%, which we
 1173 report but do not lead with, since 37 of 42 pairs are negatives and specificity is inflated by that
 1174 imbalance; precision 1.0 at base rate 11.9% is the 8.4 $\times$ enrichment that matters, and recall is 40%
 1175 under the tightened C1, since FG→RFA now counts as a miss). We report this pooled figure with
 1176 its decomposition rather than on its own, because the tiers are not equivalent evidence: it is 16/18
 1177 development *fit* plus 22/24 held-out *prediction*. Scoring instead against the criterion’s mechanism
 1178 labels, which classify both failed LOW calls as HIGH, gives TP = 3, FP = 0, TN = 37, FN = 2
 1179 (40/42); we take the more conservative pooled figure as primary. The two false positives—fg→NC
 1180 (0.927) and fedavg→CM (0.519)—were hedged as speculative at pre-registration time and neither
 1181 satisfies C1; every pair that actually satisfies C1 $\wedge$ C2 $\wedge$ C3 came out LOW. Category tally after the
 1182 Proposition 1 correction: degenerate 18, C1-fail 11, C2-fail 3, C3-fail 7, PASS 3—and all 3 surviving

1183 C2 failures are $\cdot \rightarrow$ Reputation, exactly the non-invariant class of Proposition 1(c). With a positive
 1184 set of only 2 certified in-distribution pairs and recall 40%, the 94.6% specificity figure is high partly
 1185 because the positive class is rare, so we report it as a supporting statistic rather than a headline.
 1186 The conditions are sufficient, not necessary, and they do reject working compositions. Extension to
 1187 3-way compositions: 9/10 triples correct (too small for confidence).

1188 P Diagnosing the two false negatives in full

1189 NormClip $\rightarrow$ CM (0.433) and RFA $\rightarrow$ CM (0.485) were labelled C3-FAIL on the reading that *any* het-
 1190 erogeneous upstream rescaling compresses a rank aggregator’s margin. Theorem 1(1) says some-
 1191 thing weaker and quantitative: ordering survives when the attack’s coordinate extremeness exceeds
 1192 the realized weight ratio, $S > \rho$. Measured, NormClip’s ratio is mild. Over 9 live rounds it clips
 1193 only 0–2 of 5 clients (11 of 45 client-rounds), so $\approx 76\%$ of clients receive $c_i = 1$ *exactly*, and the
 1194 largest raw norm we observe is 24.5 against $\tau=5$, bounding $\rho \leq 4.9$; against $S=10$ the margin
 1195 $S/\rho \geq 2.0$ still clears 1. RFA-reweighting is likewise near-uniform on benign clients. Both pairs
 1196 therefore sit close to the *uniform* rescaling case, where ordering is preserved exactly—which is why
 1197 their ASR lands in an intermediate band just under threshold rather than in the high band the binary
 1198 label predicts. **The refinement does not recover them.** Both also fail C1: CoordMedian alone is
 1199 already at 0.519, and neither NormClip (0.935) nor RFA (0.885) suppresses, so no reading of C3
 1200 makes either pair certifiable. What the diagnosis buys is a sharper condition and an account of the
 1201 residual: a d_2 whose own baseline sits at 0.519 needs only a mild order-preserving upstream trans-
 1202 form to cross 0.5, so these two are threshold noise around a marginal aggregator, not suppression
 1203 the framework failed to recognise. The operational refinement is to evaluate C3 for rank aggrega-
 1204 tors against the *realized* ρ from a 5–10 round pilot rather than the binary question of whether d_1 is
 1205 heterogeneous—the same quantity Theorem 1(1) already requires, which the labelling had not been
 1206 using.

1207 Q Condition-by-condition scoring, and the run-count saving

1208 The results below are §6’s, moved here for space. None changes a reported number.

1209 **The out-of-sample bookkeeping, in full.** All 4 out-of-sample PASS predictions the tightened cri-
 1210 terion licenses held, as did both in-distribution C1 $\wedge$ C2 $\wedge$ C3 pairs. **But all 7 are *positive* predictions**
 1211 **whose FAIL controls are uninformative** (DBA fails even undefended), which is why §6 declines
 1212 to read them as an out-of-sample classification result: five positive predictions against an 11.9%
 1213 base rate (5/42, Appendix O), with the positive direction separately unvalidated (§4).

1214 **Does every condition earn its place? C3 does not—reported as a finding.** Scoring all 42 pairs
 1215 on each condition *independently*: C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, C1 $\wedge$ C2
 1216 selects 2 at 100%, and **C3 alone selects 35 at 9%**—below the 12% base rate, never binding on a
 1217 pair C1 $\wedge$ C2 already admits. So **we demote C3 to a margin diagnostic and recommend C1 $\wedge$ C2**,
 1218 retaining it on theoretical grounds only: Theorem 1 makes margin compression a real failure mode
 1219 and the $\alpha=0.1$ failure of §6 is exactly a margin collapse. It changes no number: a finding about the
 1220 criterion, not an improvement to it.

1221 **What the saving is.** Counted in runs rather than pairs—C1’s standalone evaluations are runs too—
 1222 the saving is 420 **unscreened** against 80 **screened**, 81% **fewer composition-evaluation runs**, not
 1223 the 95.2% of *pairs* we previously reported; **nothing is eliminated before any run**, and the structural
 1224 claim is the scaling $|A| S D(D-1)$ (*quadratic* in the defense count) to $|A| S ((D-1) + |\text{admitted}|)$.
 1225 GPU-hour figures are *estimates*—no timing telemetry was recorded.

Table 8: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 2). Every entry is scoped to the measured setting of §5.

question	identifiable?	answer, and its evidence
Does disturbing d_2 ’s statistic change its <i>decision</i> ?	measured	Yes: Krum’s selection flips in up to 0.80 of rounds
Does it change the <i>admitted adversarial mass</i> ?	by construction	No: exactly 0.000 at every rung
Does it change <i>suppression</i> ?	instrumented	No: $ \Delta =0.026$, inside margin (Prop. 4(a))
Does <i>adversary-correlated</i> reweighting change suppression?	instrumented	Yes , non-monotonically: single-peaked (Mode A)
Does admitted mass <i>order</i> outcomes across defenses?	pre-registered	No—refuted: $+0.098$ against a frozen $+0.150$
Is (P4) <i>sufficient</i> for (P5)?	control arm	No: <code>cos_krum</code> admits 0.000 while ASR falls
Can (P4) be screened <i>without</i> adversary identity?	no—open	Both instruments are oracles: the stated open problem
Does C1 identify C2’s contribution?	no—structural	Empty cell by the gate, for every menu (Prop. 2)
Does any of this certify <i>deployed</i> security?	no—scope	Committed attack only; one adaptive test breaks it

R What the boundary permits, claim by claim

S Experimental Details

Figure 7 places the evaluated settings in the (γ, p_{eff}) plane. At the persistence we measure ($\hat{\gamma} \approx 0.66$) the one temporal mix we ran reaches ASR 0.956, which is the reason the paper composes rather than switches; the figure records those points and draws no boundary between them.

FL configuration. $N=10$ clients, $K=5$ per round, 50 rounds, SGD with momentum 0.9, learning rate 0.01, local epochs 3, batch size 32, Dirichlet $\alpha=0.5$ (non-IID). Scale experiments: $N=100$, $K=10$, $f=0.05$, ResNet18 with GroupNorm.

Attacks. Model-scaling: scale factor 10, 50% poison fraction, target class 0. Pixel backdoor: 5×5 bottom-right trigger, target class 0. DBA: 4 sub-triggers at image corners, $f=0.4/0.6$. Joint-constraint: $\epsilon \in \{0.5, 1.0, 2.0, 4.0\}$, disjoint coordinate blocks per adversary, single-step ϵ -ball projection per round.

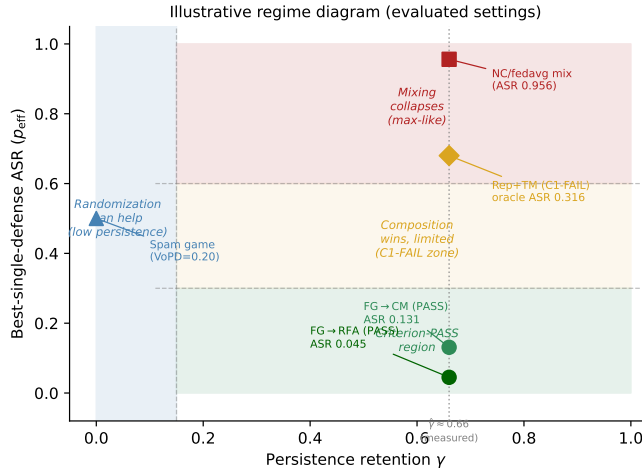


Figure 7: **The evaluated settings, plotted. No regimes, and one line, which is a measurement.** X-axis: persistence γ . Y-axis: best single-defense ASR (p_{eff}). Each marker is a setting we ran and carries its own measured value; the dotted line is the measured FL persistence $\hat{\gamma} \approx 0.66$. **An earlier version shaded four regime bands and drew threshold guides at $\gamma=0.15$, $p_{\text{eff}}=0.3$ and $p_{\text{eff}}=0.6$; none of those numbers was measured, so all six are gone.** Five points cannot locate a boundary, and nothing here is evidence of a phase transition in (γ, p_{eff}) space.

Defenses. NormClip $\tau=5.0$, i.e. $u_i \mapsto u_i \min(1, \tau/\|u_i\|_2)$ over the whole flattened update; TrimmedMean $\beta=0.2$ (trim 20% each tail); RFA: unweighted Weiszfeld, 50 iterations, tolerance 10^{-6} , initialised at the mean; FoolsGold: $w_i \propto \log((1 - s_i)/(s_i + \epsilon) + \epsilon)$ with s_i the client’s maximum absolute pairwise cosine similarity normalised by the round maximum, $\epsilon=10^{-5}$, then shifted by $-\min_i$ and normalised—so the least-trusted client of each round receives weight exactly 0 (empirically 2 of 5 clients per round).

Reputation. Despite the name this defense is *memoryless*—it keeps no cumulative per-client state and is recomputed from scratch each round from that round’s K participants. Consensus is the coordinate-wise median m of the K updates; scores are ℓ_2 distances $d_i = \|u_i - m\|_2$; the bandwidth is adaptive, $\sigma = \text{median}_i(d_i)$ (clamped $\geq 10^{-6}$); weights are $w_i = \exp(-d_i/\sigma)$, normalised to sum to 1. No clipping, no initialisation. Its discriminative property is therefore *consensus distance*, not similarity spread—which is why Proposition 1(c) predicts that $\cdot \rightarrow \text{Reputation}$ compositions fail C2. Weights are strictly positive, so Reputation never produces the zero vectors of Lemma 1.

Adversary count per round. Adversaries are the fixed client identities $\{0, \dots, fN - 1\}$; each round samples K participants uniformly without replacement from all N , with no stratification. For $N=10$, $f=0.2$, $K=5$ the per-round adversary count is hypergeometric: $P(n_a=0) = P(n_a=2) = 56/252 = 0.222$, $P(n_a=1) = 140/252 = 0.556$; measured over 100 rounds we observe 0.19/0.54/0.27. Two consequences, both matching the data. CoordMedian’s count majority $n_a < K/2 = 2.5$ holds for every attainable $n_a \in \{0, 1, 2\}$, i.e. in 100% of rounds. TrimmedMean’s budget is $\lfloor \beta K \rfloor = \lfloor 1.0 \rfloor = 1$, so it is exceeded whenever $n_a=2$ —22.2% of rounds, ≈ 11 of 50. This is why FG $\rightarrow$ TrimmedMean fails (0.508) while FG $\rightarrow$ CoordMedian succeeds (0.107) under an otherwise identical upstream transform. We retain the pre-registered C3-FAIL label for FG $\rightarrow$ TM; the count-budget violation is an additional mechanism, not a relabelling.

Theorem quantities. FoolsGold weight ratio across 5 seeds $\times$ 20 rounds ($N=10$, $f=0.2$, model-scaling): $\max \rho = 2.1$, giving extremeness ratio $S/\rho \geq 4.8$ with $S=10$ (regime A). Reputation, same protocol (5 seeds $\times$ 20 rounds, 81 rounds containing both adversarial and benign clients): ρ mean 1.44×10^5 , max 2.84×10^6 , p95 6.1×10^5 ; mean adversarial weight 4.83×10^{-4} against mean benign weight 0.278, and $w_a < 10^{-3}$ in 85% of rounds. Reputation therefore fails the ordering condition by ~ 5 orders of magnitude and suppresses via regime B instead. RFA, measured on the transformed points $u'_i = w_i u_i$ that RFA actually receives (25 adversary-present rounds over 3 seeds): FG zeroes every adversary in 32% of rounds, where Lemma 1 applies and the measured contraction $1 - \Lambda_a$ is 0.29 on average (range 0.001–0.52). In the other 68% ($w_a > 0$) the separation condition of Theorem 1(2) is satisfied in *every* round, with margin $\Delta'_{\text{sep}}/2(R'_B + \delta') \geq 2.8$ (mean 3.2) and δ'/R'_B mean 0.19, max 0.72. The two cases are exhaustive, so the analysis covers all measured rounds. Proposition 1(a) verified in the shipped implementation: FoolsGold’s weight vector changes by at most 3×10^{-8} (float32 rounding) under NormClip across 1000 synthetic trials and live rounds in which 1–2 of 5 clients were genuinely clipped (raw norms up to 24.5 against $\tau=5$). Exactly 2 of the 5 clients per round receive zero FoolsGold weight. **Corollary 1 in full:** with $D_b := R'_B + \delta'$ bounding every benign residual and $D_a := \Delta'_{\text{sep}} - R'_B - \delta' = (2m - 1)D_b$ lower-bounding every adversarial one, the Weiszfeld coefficients satisfy $\sum_j 1/d_j \geq n_b/D_b$ and $1/d_a \leq 1/D_a$, so $\Lambda_a \leq (n_a/D_a)/(n_b/D_b + n_a/D_a) = n_a/(n_b(2m - 1) + n_a)$. At $m \rightarrow 1$ this is $n_a/(n_b + n_a)$, i.e. 40% for $n_a=2$, $n_b=3$; at the measured $m \geq 2.82$ it is $\leq 12.6\%$, and at the all-round mean $m = 3.24$, $\leq 10.8\%$.

Pre-registration provenance. We distinguish the tiers explicitly. The 18-pair development set is where the criterion was *developed*; its predictions were not committed in advance and we report it as fit. The 24-pair held-out set was pre-registered in `pre_registration_wave2.md` and committed before those experiments ran; it contains no predicted-PASS pair, so it bears only on specificity. The out-of-sample PASS predictions (DBA transfer, $N=100/\text{ResNet18}$ scale transfer, composition dominance) were pre-registered separately in `pre_registration_oos_pass.md`, also committed in advance. The 10 three-way predictions were pre-registered in `pre_registration_three_way.md`. Two held-out LOW calls, hedged as speculative at pre-registration time, failed; we score them as misses rather than dropping them. The prospective suite of §5.1 was pre-registered in `pre_registration_prospective.md` at commit 262cf35, which precedes the first write to `results/prospective.suite/`; single-defense baselines for the three new defenses were

1292 measured *before* the freeze, because C1 and C0 take them as inputs (FLTrust 0.048, Krum
1293 0.583, Multi-Krum 0.783 max-committed; FedAvg 0.667 scaling / 0.768 pixel, clearing the
1294 sanity gate). The predicted labels in that file are reproduced verbatim in the PAIRS list of
1295 run_prospective_suite.py and the confusion matrix is recomputed from per-seed ASR in
1296 results/prospective_suite/summary.json, not transcribed.

1297 **Freeze commits and run tallies for the interventional suites.** Every design, grid
1298 and threshold below was git-committed before the corresponding results directory ex-
1299 isted, and each hash is verifiable in the repository history. *Dose-response lad-*
1300 *der* (§5.2): pre_registration_dose_response.md at commit e711a95, before
1301 results/dose_response/; 80 runs. *Metric-swap mirrors* (§5.1): 8 frozen labels at commit
1302 356897b. *Prospective suite* (§5.1): commit 262cf35, as above. *Targeted intervention, Modes S*
1303 *and A* (§4, §5.3): design and thresholds at commit 5130cec before results/targeted_dose/
1304 existed; 97 runs plus 25 identity runs verified bit-identical per seed (largest deviation 0.00e+0);
1305 the adversarial coefficient share was verified with no ASR involved, before any run. *Rank-*
1306 *discordant replication* (§4): pre_registration_dose_replication.md at commit 06d9bce
1307 before results/dose_replication/ existed; 15 new runs, identity rung imported bit-identical
1308 from the ladder and verified by --harness-check.